

Q1: A soft ferromagnetic material is placed in an external magnetic field. The magnetic domains :

- (A) may increase or decrease in size and change its orientation.
- (B) increase in size but no change in orientation.
- (C) have no relation with external magnetic field.
- (D) decrease in size and changes orientation.

Q1: A soft ferromagnetic material is placed in an external magnetic field. The magnetic domains :

- (A) may increase or decrease in size and change its orientation.
- (B) increase in size but no change in orientation.
- (C) have no relation with external magnetic field.
- (D) decrease in size and changes orientation.

Solution:

Soft ferromagnetic materials are materials which can be easily magnetised and demagnetised by external magnetic field. When external field is applied, the domains experiences a net torque hence change its orientation.
Hence option (a) is correct

Q2: A particle is projected with velocity v_0 along x-axis. A damping force is acting on the particle which is proportional to the square of the distance from the origin. i.e. $ma = -\alpha x^2$. The distance at which the particle stops :

(A) $\left(\frac{2mv_0}{3\alpha}\right)^{\frac{1}{3}}$

(B) $\left(\frac{3mv_0^2}{2\alpha}\right)^{\frac{1}{2}}$

(C) $\left(\frac{2mv_0^2}{3\alpha}\right)^{\frac{1}{2}}$

(D) $\left(\frac{3mv_0^2}{2\alpha}\right)^{\frac{1}{3}}$

Q2: A particle is projected with velocity v_0 along x-axis. A damping force is acting on the particle which is proportional to the square of the distance from the origin. i.e. $ma = -\alpha x^2$. The distance at which the particle stops :

(A) $\left(\frac{2mv_0}{3\alpha}\right)^{\frac{1}{3}}$

(B) $\left(\frac{3mv_0^2}{2\alpha}\right)^{\frac{1}{2}}$

(C) $\left(\frac{2mv_0^2}{3\alpha}\right)^{\frac{1}{2}}$

(D) $\left(\frac{3mv_0^2}{2\alpha}\right)^{\frac{1}{3}}$

Solution:

$$F = -\alpha x^2$$

$$ma = -\alpha x^2$$

$$a = \frac{-\alpha x^2}{m}$$

$$\frac{v dv}{dx} = -\frac{\alpha}{m} x^2$$

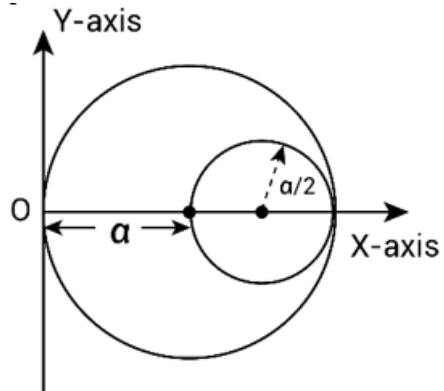
$$\int_{v_0}^0 v dv = \int_0^x -\frac{\alpha}{m} x^2 dx$$

$$\left(\frac{v^2}{2}\right)_{v_0}^0 = -\frac{\alpha}{m} \left(\frac{x^3}{3}\right)_0^x$$

$$\frac{-v_0^2}{2} = -\frac{\alpha}{m} \frac{x^3}{3}$$

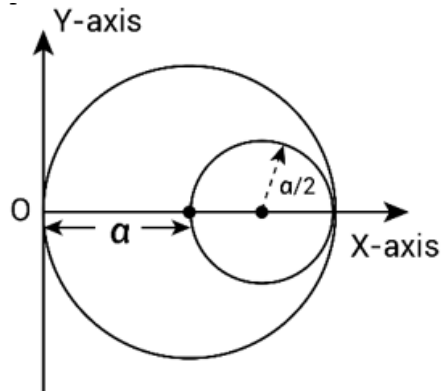
$$x = \left(\frac{3mv_0^2}{2\alpha}\right)^{\frac{1}{3}}$$

Q3: A circular hole of radius $\left(\frac{a}{2}\right)$ is cut out of a circular disc of radius 'a' as shown in figure. The centre of mass of the remaining circular portion with respect to point 'O' will be:



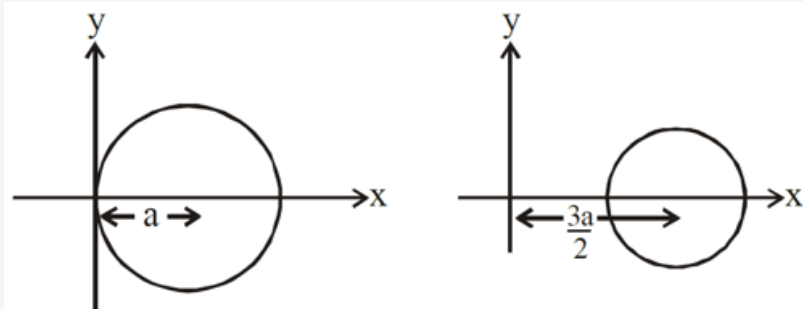
- (A) $\frac{1}{6}a$
- (B) $\frac{5}{6}a$
- (C) $\frac{2}{3}a$
- (D) $\frac{10}{11}a$

Q3: A circular hole of radius $\left(\frac{a}{2}\right)$ is cut out of a circular disc of radius 'a' as shown in figure. The centre of mass of the remaining circular portion with respect to point 'O' will be:



- (A) $\frac{1}{6}a$
- (B) $\frac{5}{6}a$
- (C) $\frac{2}{3}a$
- (D) $\frac{10}{11}a$

Solution:



Let σ be the uniform mass density of disc then

$$x_{COM} = \frac{(\sigma\pi a^2) a - \sigma\pi \left(\frac{a^2}{4}\right) \times \frac{3a}{2}}{\sigma\pi^2 a - \frac{\sigma\pi a^2}{4}}$$

$$= \frac{a - \frac{3a}{8}}{1 - \frac{1}{4}} = \frac{5a}{6}$$

Q4: Match List - I with List - II.

List - I	List - II
(a) Source of microwave frequency	(i) Radioactive decay of nucleus
(b) Source of infrared frequency	(ii) Magnetron
(c) Source of Gamma Rays	(iii) Inner shell electrons
(d) Source of X-rays	(iv) Vibration of atoms and molecules
	(v) LASER
	(vi) RC circuit

Choose the correct answer from the options given below :

- (A) (a) – (ii), (b) – (iv), (c) – (i), (d) – (iii)
- (B) (a) – (vi), (b) – (iv), (c) – (i), (d) – (v)
- (C) (a) – (vi), (b) – (v), (c) – (i), (d) – (iv)
- (D) (a) – (ii), (b) – (iv), (c) – (vi), (d) – (iii)

Q4: Match List - I with List - II.

List – I	List - II
(a) Source of microwave frequency	(i) Radioactive decay of nucleus
(b) Source of infrared frequency	(ii) Magnetron
(c) Source of Gamma Rays	(iii) Inner shell electrons
(d) Source of X-rays	(iv) Vibration of atoms and molecules
	(v) LASER
	(vi) RC circuit

Choose the correct answer from the options given below :

- (A) (a) – (ii), (b) – (iv), (c) – (i), (d) – (iii)
- (B) (a) – (vi), (b) – (iv), (c) – (i), (d) – (v)
- (C) (a) – (vi), (b) – (v), (c) – (i), (d) – (iv)
- (D) (a) – (ii), (b) – (iv), (c) – (vi), (d) – (iii)

Solution:

- (a) Source of microwave frequency is magnetron.
- (b) Source of infrared frequency is vibration of atoms and molecules
- (c) Source of Gamma rays is radioactive decay of nucleus
- (d) Source of X- rays inner shell electron transition.

Q5: Two electrons each are fixed at a distance '2d'. A third charge proton placed at the midpoint is displaced slightly by a distance x ($x \ll d$) perpendicular to the line joining the two fixed charges. Proton will execute simple harmonic motion having angular frequency : (m = mass of charged particle)

(A) $\left(\frac{q^2}{2\pi\epsilon_0 m d^3} \right)^{\frac{1}{2}}$

(B) $\left(\frac{2\pi\epsilon_0 m d^3}{q^2} \right)^{\frac{1}{2}}$

(C) $\left(\frac{2q^2}{\pi\epsilon_0 m d^3} \right)^{\frac{1}{2}}$

(D) $\left(\frac{\pi\epsilon_0 m d^3}{2q^2} \right)^{\frac{1}{2}}$

Q5: Two electrons each are fixed at a distance '2d'. A third charge proton placed at the midpoint is displaced slightly by a distance x ($x \ll d$) perpendicular to the line joining the two fixed charges. Proton will execute simple harmonic motion having angular frequency : (m = mass of charged particle)

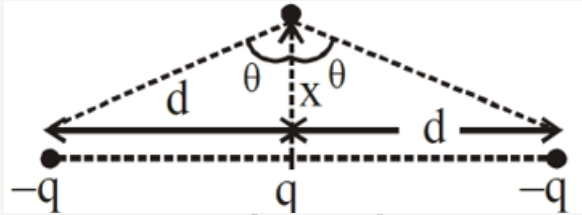
(A) $\left(\frac{q^2}{2\pi\epsilon_0 m d^3} \right)^{\frac{1}{2}}$

(B) $\left(\frac{2\pi\epsilon_0 m d^3}{q^2} \right)^{\frac{1}{2}}$

(C) $\left(\frac{2q^2}{\pi\epsilon_0 m d^3} \right)^{\frac{1}{2}}$

(D) $\left(\frac{\pi\epsilon_0 m d^3}{2q^2} \right)^{\frac{1}{2}}$

Solution:



$$F_{netq} = - [2F_{q/q} \cos \theta]$$

$$F_{netq} = -2 \cdot \frac{1}{4\pi\epsilon_0} \cdot \frac{q^2}{(\sqrt{d^2 + x^2})^2} \cdot \frac{x}{\sqrt{d^2 + x^2}}$$

$$= -\frac{q^2}{2\pi\epsilon_0} \frac{x}{(d^2 + x^2)^{3/2}}$$

For $x \ll d$,

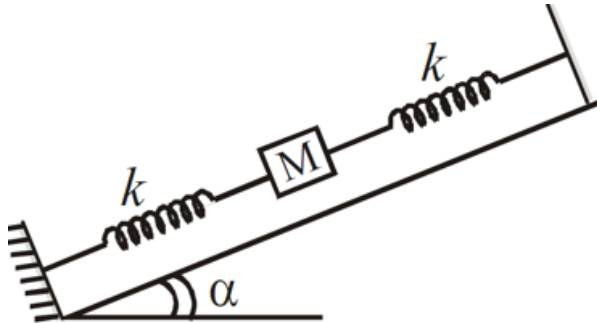
$$F_{net\ q} = -\frac{q^2}{2\pi\epsilon_0 d^3} x$$

$$\therefore a = -\frac{q^2}{2\pi\epsilon_0 \cdot m d^3} x$$

Comparing with equation of SHM ($a = -\omega^2 x$)

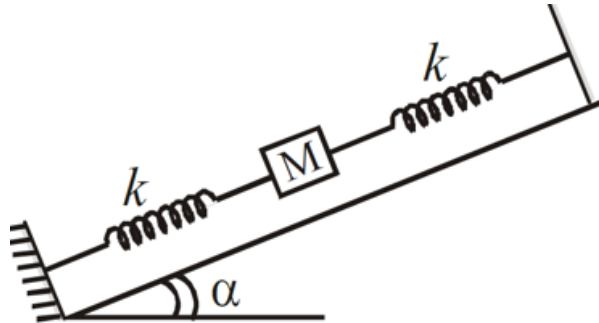
$$\therefore \omega = \sqrt{\frac{q^2}{2\pi\epsilon_0 m d^3}}$$

Q6: In the given figure, a body of mass M is held between two massless springs, on a smooth inclined plane. The free ends of the springs are attached to firm supports. If each spring has spring constant k , the frequency of oscillation of given body is :



- (A) $\frac{1}{2\pi} \sqrt{\frac{2k}{M}}$
- (B) $\frac{1}{2\pi} \sqrt{\frac{k}{2M}}$
- (C) $\frac{1}{2\pi} \sqrt{\frac{2k}{Mg \sin \alpha}}$
- (D) $\frac{1}{2\pi} \sqrt{\frac{k}{Mg \sin \alpha}}$

Q6: In the given figure, a body of mass M is held between two massless springs, on a smooth inclined plane. The free ends of the springs are attached to firm supports. If each spring has spring constant k , the frequency of oscillation of given body is :



- (A) $\frac{1}{2\pi} \sqrt{\frac{2k}{M}}$
- (B) $\frac{1}{2\pi} \sqrt{\frac{k}{2M}}$
- (C) $\frac{1}{2\pi} \sqrt{\frac{2k}{Mg \sin \alpha}}$
- (D) $\frac{1}{2\pi} \sqrt{\frac{k}{Mg \sin \alpha}}$

Solution:

$$k_{eq} = k_1 + k_2 = k + k = 2k$$

$$T = 2\pi \sqrt{\frac{m}{k_{eq}}} = 2\pi \sqrt{\frac{m}{2k}}$$

$$f = \frac{1}{T} = \frac{1}{2\pi} \sqrt{\frac{2k}{m}}$$

Q7: The period of oscillation of a simple pendulum is $T = 2\pi\sqrt{\frac{L}{g}}$. Measured value of 'L' is 1.0 m from meter scale having a minimum division of 1 mm and time of one complete oscillation is 1.95 s measured from stopwatch of 0.01s resolution. The percentage error in the determination of 'g' will be:

- (A) 1.03%
- (B) 1.33%
- (C) 1.13%
- (D) 1.30%

Q7: The period of oscillation of a simple pendulum is $T = 2\pi\sqrt{\frac{L}{g}}$. Measured value of 'L' is 1.0 m from meter scale having a minimum division of 1 mm and time of one complete oscillation is 1.95 s measured from stopwatch of 0.01s resolution. The percentage error in the determination of 'g' will be:

- (A) 1.03%
- (B) 1.33%
- (C) 1.13%
- (D) 1.30%

Solution:

$$T = 2\pi\sqrt{\frac{\ell}{g}}$$

$$g = \frac{4\pi^2 \ell}{T^2}$$

$$\frac{\Delta g}{g} = \frac{\Delta \ell}{\ell} + \frac{2\Delta T}{T}$$

$$\frac{\Delta g}{g} = \frac{1 \times 10^{-3}}{1} + 2 \times \frac{0.01}{1.95}$$

$$\frac{\Delta g}{g} = 0.013 \text{ or } 1.13\%$$

Q8: The de Broglie wavelength of a proton and α -particle are equal. The ratio of their velocities is

- (A) 4 : 2
- (B) 4 : 3
- (C) 1 : 4
- (D) 4 : 1

Q8: The de Broglie wavelength of a proton and α -particle are equal. The ratio of their velocities is

(A) 4 : 2

(B) 4 : 3

(C) 1 : 4

(D) 4 : 1

Solution:

$$\lambda = \frac{h}{mv}$$

$$\lambda_P = \lambda_\alpha$$

$$m_P v_P = m_\alpha v_\alpha$$

$$m_P v_P = 4m_P v_\alpha \quad (m_\alpha = 4m_P)$$

$$\frac{v_P}{v_\alpha} = 4$$

Q9: Given below are two statements

Statement I : PN junction diodes can be used to function as transistors, simply by connecting two diodes, back to back, which act as the base terminal.

Statement II : In the study of transistor the amplification factor β indicates ratio of the collector current to the base current.

In the light of the above statements, choose the correct answer from the options given below

- (A) Statement I is false but Statement II is true
- (B) Both Statement I and Statement II are true
- (C) Both Statement I and Statement II are false
- (D) Statement I is true but Statement II is false

Q9: Given below are two statements

Statement I : PN junction diodes can be used to function as transistors, simply by connecting two diodes, back to back, which act as the base terminal.

Statement II : In the study of transistor the amplification factor β indicates ratio of the collector current to the base current.

In the light of the above statements, choose the correct answer from the options given below

- (A) Statement I is false but Statement II is true
- (B) Both Statement I and Statement II are true
- (C) Both Statement I and Statement II are false
- (D) Statement I is true but Statement II is false

Solution:

Back to back diode will not make a transistor.

$$\beta = \frac{i_c}{i_b}$$

Q10: An X-ray tube is operated at 1.24 million volt. The shortest wavelength of the produced photon will be

- (A) 10^{-3} nm
- (B) 10^{-2} nm
- (C) 10^{-4} nm
- (D) 10^{-1} nm

Q10: An X-ray tube is operated at 1.24 million volt. The shortest wavelength of the produced photon will be

- (A) 10^{-3} nm
- (B) 10^{-2} nm
- (C) 10^{-4} nm
- (D) 10^{-1} nm

Solution:

$$\begin{aligned}\lambda_{\min} &= \frac{1240}{\Delta V} (nm) \\ &= \frac{1240}{1.24 \times 10^6} = 10^{-3} nm\end{aligned}$$

Q11: A body weighs 49 N on a spring balance at the north pole. What will be its weight recorded on the same weighing machine, if it is shifted to the equator ?

[use $g = \frac{GM}{R^2} = 9.8\text{ms}^{-2}$ and radius of earth, $R = 6400 \text{ km}$.]

- (A) 49.17 N
- (B) 48.83 N
- (C) 49 N
- (D) 49.83 N

Q11: A body weighs 49 N on a spring balance at the north pole. What will be its weight recorded on the same weighing machine, if it is shifted to the equator ?

[use $g = \frac{GM}{R^2} = 9.8\text{ms}^{-2}$ and radius of earth, $R = 6400 \text{ km}$.]

(A) 49.17 N

(B) 48.83 N

(C) 49 N

(D) 49.83 N

Solution:

Weight at pole = $mg = 49 \text{ N}$

At equator due to rotation = $g_e = g - R\omega^2$

$$W = mg_e = m(g - R\omega^2)$$

$$\therefore W_p > W_e \quad W_p = 49\text{N}$$

$$\text{So, } W_e = 48.88 \text{ N.} \quad W_e < 49\text{N}$$

Option (b) is correct.

Q12: Which of the following equations represents a travelling wave?

(A) $y = A \sin(15x - 2t)$

(B) $y = Ae^{-x^2} (vt + \theta)$

(C) $y = Ae^x \cos(\omega t - \theta)$

(D) $y = A \sin x \cos \omega t$

Q12: Which of the following equations represents a travelling wave?

(A) $y = A \sin(15x - 2t)$

(B) $y = Ae^{-x^2} (vt + \theta)$

(C) $y = Ae^x \cos(\omega t - \theta)$

(D) $y = A \sin x \cos \omega t$

Solution:

$$y = F(x, t)$$

For travelling wave y should be linear function of x and t and they must exist as $(x \pm vt)$

$y = \sin(15x - 2t) \rightarrow$ linear function in x and t .

So, option (a) is correct.

Q13: According to Bohr atom model, in which of the following transitions will the frequency be maximum ?

(A) $n = 3$ to $n = 2$

(B) $n = 2$ to $n = 1$

(C) $n = 5$ to $n = 4$

(D) $n = 4$ to $n = 3$

Q13: According to Bohr atom model, in which of the following transitions will the frequency be maximum ?

(A) $n = 3$ to $n = 2$

(B) $n = 2$ to $n = 1$

(C) $n = 5$ to $n = 4$

(D) $n = 4$ to $n = 3$

Solution:

$$\Delta E = 13.6 \left[\frac{1}{n_1^2} - \frac{1}{n_2^2} \right] = h\nu$$

It is maximum if $n_1 = 1$ and $n_2 = 2$

$n = 5$-0.55 eV

$n = 4$-0.850 eV

$n = 3$-1.511 eV

$n = 2$-3.4 eV

$n = 1$-13.6 eV

Option (b) is correct.

Q14: On the basis of kinetic theory of gases, the gas exerts pressure because its molecules :

- (A) continuously lose their energy till it reaches wall.
- (B) are attracted by the walls of container
- (C) continuously stick to the walls of container.
- (D) suffer change in momentum when impinge on the walls of container.

Q14: On the basis of kinetic theory of gases, the gas exerts pressure because its molecules :

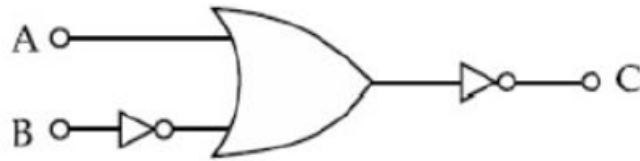
- (A) continuously lose their energy till it reaches wall.
- (B) are attracted by the walls of container
- (C) continuously stick to the walls of container.
- (D) suffer change in momentum when impinge on the walls of container.

Solution:

From the assumption of KTG, the molecules of gas collide with the walls and suffers momentum change which results in force on the wall and hence pressure.

Hence option (d) is correct.

Q15:



The logic circuit shown above is equivalent to

- (A)

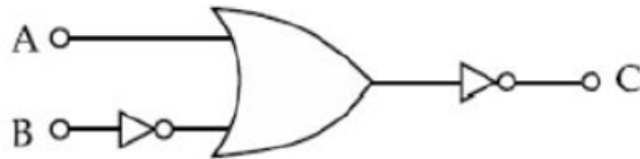
A logic circuit diagram showing an AND gate. The top input is labeled 'A' and passes through an inverter before entering the AND gate. The bottom input is labeled 'B'. The output of the AND gate is labeled 'C'.
- (B)

A logic circuit diagram showing an AND gate. The top input is labeled 'A'. The bottom input is labeled 'B' and passes through an inverter before entering the AND gate. The output of the AND gate is labeled 'C'.
- (C)

A logic circuit diagram showing an OR gate. The top input is labeled 'A'. The bottom input is labeled 'B'. The output of the OR gate is labeled 'C'.
- (D)

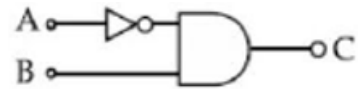
A logic circuit diagram showing an AND gate. The top input is labeled 'A' and passes through an inverter before entering the AND gate. The bottom input is labeled 'B'. The output of the AND gate is labeled 'C'.

Q15:

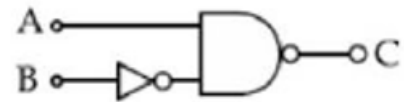


The logic circuit shown above is equivalent to

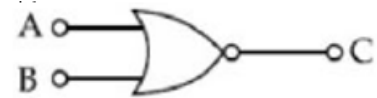
(A)



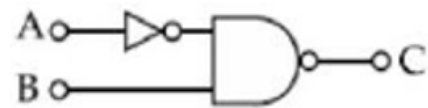
(B)



(C)



(D)



Solution:

Truth table of the given gate:

A	B	C
0	0	0
0	1	1
1	0	0
1	1	0

Truth table of option (a)

A	B	C
0	0	0
0	1	1
1	0	0
1	1	1

Truth table of option (b)

A	B	C
0	0	1
0	1	0
1	0	1
1	1	1

Truth table of option (c)

A	B	C
0	0	1
0	1	0
1	0	0
1	1	0

Truth table of option (d)

A	B	C

0	0	1
0	1	1
1	0	0
1	1	0

Since option (a) has same truth table, hence answer is option (a).

Q16: When a particle executes SHM, the nature of graphical representation of velocity as a function of displacement

- (A) parabolic
- (B) straight line
- (C) circular
- (D) elliptical

Q16: When a particle executes SHM, the nature of graphical representation of velocity as a function of displacement

- (A) parabolic
- (B) straight line
- (C) circular
- (D) elliptical

Solution:

For a particle executing SHM,

$$x = A \sin(\omega t + \phi)$$

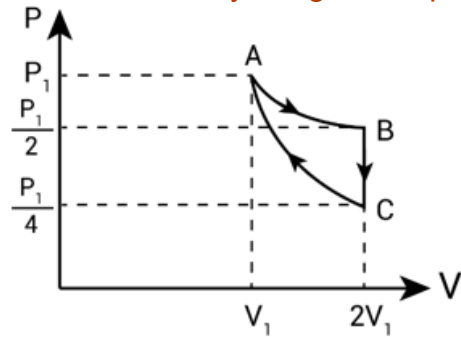
$$v = \omega A \cos(\omega t + \phi)$$

$$\Rightarrow \frac{v^2}{\omega^2 A^2} + \frac{x^2}{A^2} = 1$$

$\Rightarrow$ Equation of ellipse between v and x

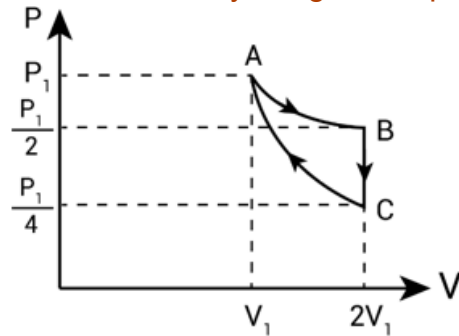
Hence option (d).

Q17: If one mole of an ideal gas at (P_1, V_1) is allowed to expand reversibly and isothermally (A to B) its pressure is reduced to one-half of the original pressure (see figure). This is followed by a constant volume cooling till its pressure is reduced to one-fourth of the initial value ($B \rightarrow C$). Then it is restored to state by a reversible adiabatic compression ($C \rightarrow A$). The net workdone by the gas is equal to



- (A) $RT \left(\ln 2 - \frac{1}{2(\gamma-1)} \right)$
- (B) $-\frac{RT}{2(\gamma-1)}$
- (C) $RT \ln 2$
- (D) 0

Q17: If one mole of an ideal gas at (P_1, V_1) is allowed to expand reversibly and isothermally (A to B) its pressure is reduced to one-half of the original pressure (see figure). This is followed by a constant volume cooling till its pressure is reduced to one-fourth of the initial value ($B \rightarrow C$). Then it is restored to state by a reversible adiabatic compression ($C \rightarrow A$). The net workdone by the gas is equal to



(A) $RT \left(\ln 2 - \frac{1}{2(\gamma-1)} \right)$

(B) $-\frac{RT}{2(\gamma-1)}$

(C) $RT \ln 2$

(D) 0

Solution:

A – B = isothermal process

$$W_{AB} = P_1 V_1 \ln\left(\frac{2V_1}{V_1}\right) = P_1 V_1 \ln(2)$$

B – C → Isochoric process

$$W_{BC} = 0$$

C – A → Adiabatic process

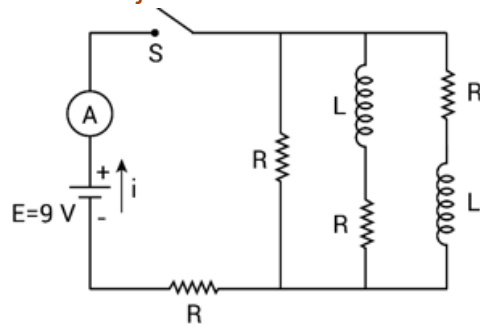
$$W_{CA} = \frac{P_1 V_1 - \frac{P_1}{4} \times 2V_1}{1 - \gamma} = \frac{P_1 V_1 \left[1 - \frac{1}{2}\right]}{1 - \gamma} = \frac{P_1 V_1}{2(1 - \gamma)}$$

$$W_{net} = W_{AB} + W_{BC} + W_{CA} \quad \{P_1 V_1 = RT\}$$

$$= P_1 V_1 \ln(2) + 0 + \frac{P_1 V_1}{2(1 - \gamma)}$$

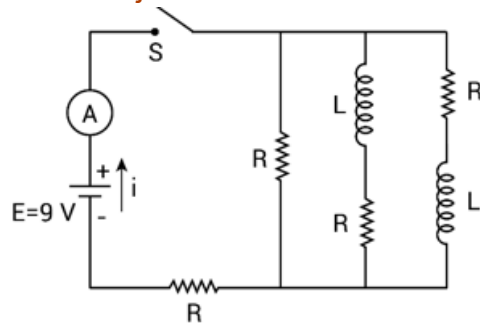
$$W_{net} = RT \left[\ln(2) - \frac{1}{2(\gamma - 1)} \right]$$

Q18: Figure shows a circuit that contains four identical resistors with resistance $R = 2.0 \, \Omega$, two identical inductors with inductance $L = 2.0 \, \text{mH}$ and an ideal battery, with emf $E = 9 \, \text{V}$. The current ' i ' just after the switch ' S ' is closed will be:



- (A) 3.0 A
- (B) 9 A
- (C) 3.37 A
- (D) 2.25 A

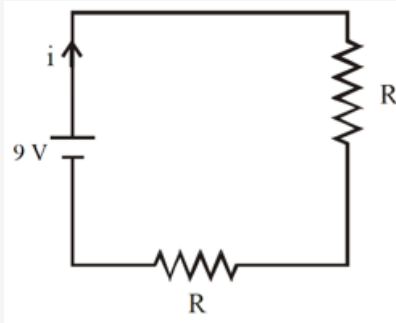
Q18: Figure shows a circuit that contains four identical resistors with resistance $R = 2.0 \, \Omega$, two identical inductors with inductance $L = 2.0 \, \text{mH}$ and an ideal battery, with emf $E = 9 \, \text{V}$. The current ' i ' just after the switch ' S ' is closed will be:



- (A) 3.0 A
- (B) 9 A
- (C) 3.37 A
- (D) 2.25 A

Solution:

Just after the switch is closed, inductor will behave like infinite resistance (open circuit) so the circuit will look like



$$i = \frac{9}{R+R} = \frac{9}{4} = 2.25\text{A}$$

Option (d) is correct

Q19: Zener breakdown occurs in a p - n junction having p and n both :

- (A) lightly doped and have wide depletion layer.
- (B) heavily doped and have wide depletion layer.
- (C) lightly doped and have narrow depletion layer.
- (D) heavily doped and have narrow depletion layer.

Q19: Zener breakdown occurs in a p - n junction having p and n both :

- (A) lightly doped and have wide depletion layer.
- (B) heavily doped and have wide depletion layer.
- (C) lightly doped and have narrow depletion layer.
- (D) heavily doped and have narrow depletion layer.

Solution:

Zener diode is heavily doped and have narrow depletion layer.

Q20: If the source of light used in a Young's double slit experiment is changed from red to violet:

- (A) the fringes will become brighter.
- (B) the intensity of minima will increase.
- (C) the central bright fringe will become a dark fringe.
- (D) consecutive fringe lines will come closer.

Q20: If the source of light used in a Young's double slit experiment is changed from red to violet:

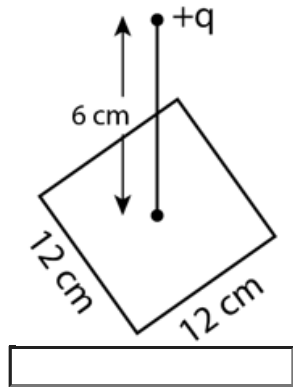
- (A) the fringes will become brighter.
- (B) the intensity of minima will increase.
- (C) the central bright fringe will become a dark fringe.
- (D) consecutive fringe lines will come closer.

Solution:

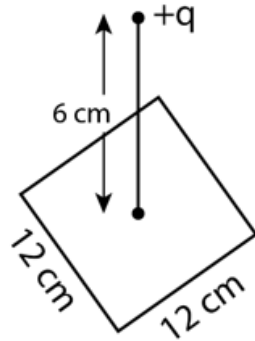
$$\beta = \frac{\lambda D}{d}$$
$$\lambda_R > \lambda_V$$
$$\beta_R = \frac{\lambda_R D}{d} \text{ and } \beta_v = \frac{\lambda_v D}{d}$$
$$\beta_R > \beta_v$$

Fringe pattern will shrink.

Q21: A point charge of $+12\ \mu\text{C}$ is at a distance 6 cm vertically above the centre of a square of side 12 cm as shown in figure. The magnitude of the electric flux through the square will be _____ $\times 10^3\ \text{Nm}^2/\text{C}$.



Q21: A point charge of $+12 \mu\text{C}$ is at a distance 6 cm vertically above the centre of a square of side 12 cm as shown in figure. The magnitude of the electric flux through the square will be _____ $\times 10^3 \text{ Nm}^2/\text{C}$.



226.00

Solution:

$$\begin{aligned}
 \text{From symmetry } \phi &= \frac{1}{6} \left(\frac{q}{\epsilon_0} \right) \\
 &= \frac{12 \times 10^{-6}}{6 \times 8.85 \times 10^{-12}} \\
 &= 225.98 \times 10^3 \frac{\text{Nm}^2}{\text{C}} \\
 &\simeq 226 \times 10^3 \frac{\text{Nm}^2}{\text{C}}
 \end{aligned}$$

Q22: A cylindrical wire of radius 0.5 mm and conductivity 5×10^7 S/m is subjected to an electric field of 10 mV/m. The expected value of current in the wire will be $x^3\pi$ mA. The value of x is _____

Q22: A cylindrical wire of radius 0.5 mm and conductivity $5 \times 10^7 \text{ S/m}$ is subjected to an electric field of 10 mV/m. The expected value of current in the wire will be $x^3\pi \text{ mA}$. The value of x is _____

5.00

Solution:

Conductivity $\sigma = 5 \times 10^7 \text{ S/m}$

Radius $r = 0.5 \text{ mm} = 5 \times 10^{-4} \text{ m}$

$$E = 10 \times 10^{-3} \frac{\text{V}}{\text{m}}$$

$$J = \sigma E = 10 \times 10^{-3} \times 5 \times 10^7$$

$$J = 5 \times 10^5$$

$$\frac{i}{A} = 5 \times 10^5$$

$$i = 5 \times 10^5 \times \pi r^2$$

$$= 5 \times 10^5 \times \pi \times (5 \times 10^{-4})^2$$

$$= 125\pi \times 10^{-3} \text{ Amp}$$

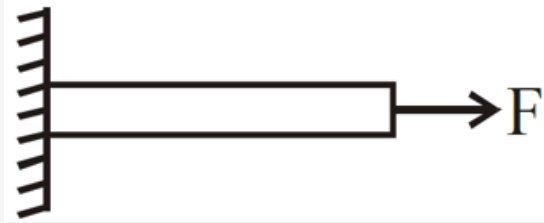
$$i = 125\pi \text{ mA}$$

$$x = 5$$

Q23: A uniform metallic wire is elongated by 0.04 m when subjected to a linear force F . The elongation, if its length and diameter is doubled and subjected to the same force will be _____ cm.

Q23: A uniform metallic wire is elongated by 0.04 m when subjected to a linear force F . The elongation, if its length and diameter is doubled and subjected to the same force will be _____ cm.

Solution:



$$F = Y \cdot A \cdot \frac{\Delta \ell}{\ell}$$

$$\Delta \ell = \frac{F}{Y \cdot A} \cdot \ell$$

$$\Delta \ell = \frac{F \cdot \ell}{T \cdot \pi r^2}$$

$$\Delta \ell \propto \frac{\ell}{r^2}$$

$$\frac{\Delta \ell_2}{\Delta \ell_1} = \left(\frac{\ell_2}{\ell_1} \right) \left(\frac{r_1}{r_2} \right)^2$$

$$= (2) \left(\frac{1}{2} \right)^2$$

$$\frac{\Delta \ell_2}{\Delta \ell_1} = \frac{1}{2}$$

$$\Delta \ell_2 = \frac{\Delta \ell_1}{2}$$

$$= \frac{0.04}{2}$$

$$= 0.02m$$

$$\boxed{\Delta \ell_2 = 2cm}$$

$$\boxed{\text{Ans.} = 2}$$

Q24: A uniform thin bar of mass 6 kg and length 2.4 m is bent to make an equilateral hexagon. The moment of inertia about an axis passing through the centre of mass and perpendicular to the plane of hexagon is $\dots \times 10^{-1} \text{ kg m}^2$.

Q24: A uniform thin bar of mass 6 kg and length 2.4 m is bent to make an equilateral hexagon. The moment of inertia about an axis passing through the centre of mass and perpendicular to the plane of hexagon is $\dots \times 10^{-1} \text{ kg m}^2$.

Solution:



m = mass of one side of hexagon = 1 kg

$$6l = 2.4$$

$$l = 0.4m$$

$$\sin 60^\circ = \frac{x}{l}$$

$$r = l \sin 60^\circ = \frac{l\sqrt{3}}{2}$$

$$\text{MOI, } I = \left[\frac{ml^2}{12} + mr^2 \right] 6$$

$$= \left[\frac{ml^2}{12} + m \left(\frac{l\sqrt{3}}{2} \right)^2 \right] 6$$

$$= 5ml^2$$

$$= 5 \times 1 \times 0.16$$

$$= 0.8$$

$$I = 8 \times 10^{-1} \text{ kg m}^2$$

Q25: An electromagnetic wave of frequency 3 GHz enters a dielectric medium of relative electric permittivity 2.25 from vacuum. The wavelength of this wave in that medium will be _____ $\times 10^{-2}$ cm.

Q25: An electromagnetic wave of frequency 3 GHz enters a dielectric medium of relative electric permittivity 2.25 from vacuum. The wavelength of this wave in that medium will be _____ $\times 10^{-2}$ cm.

667.00

Solution:

$$\lambda_{\text{in vacuum}} = \frac{c}{f} = \frac{3 \times 10^8}{3 \times 10^9} = 0.1m$$

$$\therefore \lambda_{\text{in medium}} = \frac{0.1}{\mu}$$

Where refractive index

$$\mu = \sqrt{\mu_r \epsilon_r}$$

Assuming non-magnetic material $\mu_r = 1$

$$\therefore \mu = \sqrt{2.25} = 1.5$$

$$\lambda_m = \frac{0.1}{1.5} = \frac{1}{15}m = 6.67cm$$
$$= 667 \times 10^{-2}cm$$

Ans. 667

Q26: A signal of 0.1 kW is transmitted in a cable. The attenuation of cable is -5 dB per km and cable length is 20 km. The power received at receiver is 10^{-x} W. The value of x is ____.

[Gain in dB = $10\log_{10}\left(\frac{P_0}{P_i}\right)$]

Q26: A signal of 0.1 kW is transmitted in a cable. The attenuation of cable is -5 dB per km and cable length is 20 km. The power received at receiver is 10^{-x} W. The value of x is ____.

$$[\text{Gain in dB} = 10\log_{10}\left(\frac{P_0}{P_i}\right)]$$

Solution:

Sound level decreases by 5dB every km so sound level decreased in 20 km = 100 dB

$$\beta_2 - \beta_1 = 10\log_{10}\frac{I_2}{I_1}$$

$$-100 = 10\log_{10}\frac{I_2}{I_1} \Rightarrow \frac{I_1}{I_2} = 10^{10}$$

$$I_2 = 10^{-10} I_1 \Rightarrow P_2 = 10^{-10} P_1 = 10^{-8} W$$

$$x = 8$$

Q27: The root mean square speed of molecules of a given mass of a gas at 27°C and 1 atmosphere pressure is 200 ms^{-1} . The root mean square speed of molecules of the gas at 127°C and 2 atmosphere pressure is $\frac{x}{\sqrt{3}}\text{ ms}^{-1}$. The value of x will be

Q27: The root mean square speed of molecules of a given mass of a gas at 27° C and 1 atmosphere pressure is 200 ms⁻¹. The root mean square speed of molecules of the gas at 127° C and 2 atmosphere pressure is $\frac{x}{\sqrt{3}}$ ms⁻¹. The value of x will be

400.00

Solution:

$$v_{rms} = \sqrt{\frac{3RT}{M}}$$

$$v_{rms} \propto \sqrt{T}$$

$$\frac{(v_{rms})_2}{(v_{rms})_1} = \sqrt{\frac{T_2}{T_1}}$$

$$= \sqrt{\frac{400}{300}}$$

$$= \frac{2}{\sqrt{3}}$$

$$(v_{rms})_2 = \frac{2}{\sqrt{3}}(v_{rms})_1$$

$$= \frac{2}{\sqrt{3}} \times 200$$

$$(v_{rms})_2 = \frac{400}{\sqrt{3}} \text{ m/s}$$

Ans. 400

Q28: A series LCR circuit is designed to resonate at an angular frequency $\omega_0 = 10^5$ rad/s. The circuit draws 16 W power from 120 V source at resonance. The value of resistance 'R' in the circuit is..... Ω .

Q28: A series LCR circuit is designed to resonate at an angular frequency $\omega_0 = 10^5$ rad/s. The circuit draws 16 W power from 120 V source at resonance. The value of resistance 'R' in the circuit is..... Ω .

Solution:

At resonance

$$P = \frac{V^2}{R}$$

$$R = \frac{V^2}{P} = \frac{(120)^2}{16} = 900\Omega$$

Q29: Two solids A and B of mass 1 kg and 2 kg respectively are moving with equal linear momentum. The ratio of then kinetic energies $(KE)_A : (KE)_B$ will be $\frac{A}{1}$ so the value of A will be

Q29: Two solids A and B of mass 1 kg and 2 kg respectively are moving with equal linear momentum. The ratio of then kinetic energies $(KE)_A : (KE)_B$ will be $\frac{A}{1}$ so the value of A will be

Solution:

Kinetic energy $K = \frac{P^2}{2m}, (P_A = P_B)$

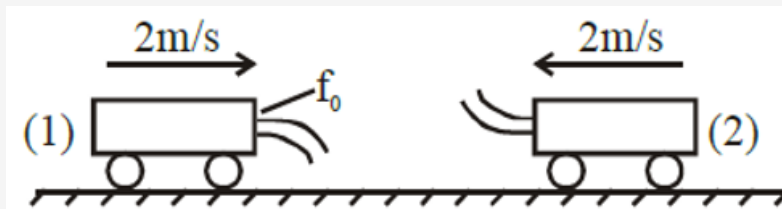
$$\begin{aligned} K &\propto \frac{1}{m} \\ \frac{K_A}{K_B} &= \frac{m_B}{m_A} \\ &= \frac{2}{1} \end{aligned}$$

Q30: Two cars are approaching each other at an equal speed of 7.2 km/hr. When they see each other, both blow horn having frequency of 676 Hz. The beat frequency heard by each driver will be _____ Hz [Velocity of sound in air is 340 m/s.]

Q30: Two cars are approaching each other at an equal speed of 7.2 km/hr. When they see each other, both blow horn having frequency of 676 Hz. The beat frequency heard by each driver will be _____ Hz [Velocity of sound in air is 340 m/s.]

8.00

Solution:



Frequency of sound heard by car -1, which comes by reflection from car-2

$$f_1 = f_0 \left(\frac{340+2}{340-2} \right) \left(\frac{340+2}{340-2} \right)$$

$$= f_0 \left(\frac{342}{338} \right)^2$$

Frequency of sound coming directly from car -2

$$f_2 = f_0 \left(\frac{340+2}{340-2} \right)$$

$$\therefore f_1 - f_2 = f_0 \left(\frac{342}{338} \right) \left(\frac{342}{338} - 1 \right) = 8.09 \simeq 8$$

Q31: Given below are two statements: one is labelled as Assertion A and the other is labelled as Reason R.

Assertion A: Hydrogen is the most abundant element in the Universe, but it is not the most abundant gas

Reason R: Hydrogen is the lightest element.

In the light of the above statements, choose the correct answer from the options given below:

- (A) Both A and R are true and R is the correct explanation of A
- (B) Both A and R are true but R is NOT the correct explanation of A
- (C) A is false but R is true
- (D) A is true but R is false

Q31: Given below are two statements: one is labelled as Assertion A and the other is labelled as Reason R.

Assertion A: Hydrogen is the most abundant element in the Universe, but it is not the most abundant gas

Reason R: Hydrogen is the lightest element.

In the light of the above statements, choose the correct answer from the options given below:

- (A) Both A and R are true and R is the correct explanation of A
- (B) Both A and R are true but R is NOT the correct explanation of A
- ☒ (C) A is false but R is true
- (D) A is true but R is false

Solution:

The most abundant gas in the troposphere is nitrogen.

Q32: Match List - I with List – II.

List – I (Metal)	List-II (Ores)
(a) Aluminium	(i) Siderite
(b) Iron	(ii) Calamine
(c) Copper	(iii) Kaolinite
(d) Zinc	(iv) Malachite

Choose the correct answer from the options given below :

- (A) (a) – (ii); (b) - (iv); (c) - (i); (d) – (iii)
- (B) (a) – (iii); (b) - (i); (c) - (iv); (d) – (ii)
- (C) (a) – (i); (b) - (iii); (c) - (iv); (d) – (ii)
- (D) (a) – (iv); (b) - (iv); (c) - (ii); (d) – (i)

Q32: Match List - I with List – II.

List – I (Metal)	List-II (Ores)
(a) Aluminium	(i) Siderite
(b) Iron	(ii) Calamine
(c) Copper	(iii) Kaolinite
(d) Zinc	(iv) Malachite

Choose the correct answer from the options given below :

- (A) (a) – (ii); (b) - (iv); (c) - (i); (d) – (iii)
- (B) (a) – (iii); (b) - (i); (c) - (iv); (d) – (ii)
- (C) (a) – (i); (b) - (iii); (c) - (iv); (d) – (ii)
- (D) (a) – (iv); (b) - (iv); (c) - (ii); (d) – (i)

Solution:

Siderite = Fe_2O_3

Malachite = $CuCO_3 \cdot Cu(OH)_2$

Calamine = $ZnCO_3$

Kaolinite = Al_2O_3

Q33: According to Bohr's atomic theory

(A) Kinetic energy of electron is $\propto \frac{Z^2}{n^2}$

(B) The product of velocity (v) of electron and principal quantum number (n), 'vn' $\propto Z^2$

(C) Frequency of revolution of electron in an orbit is $\propto \frac{Z^3}{n^3}$.

(D) Coulombic force of attraction on the electron is $\frac{Z^3}{n^4}$.

Choose the most appropriate answer from the options given below

(A) (A) only

(B) (C) only

(C) (A), (C) and (D) only

(D) (A) and (D) only

Q33: According to Bohr's atomic theory

(A) Kinetic energy of electron is $\propto \frac{Z^2}{n^2}$

(B) The product of velocity (v) of electron and principal quantum number (n), 'vn' $\propto Z^2$

(C) Frequency of revolution of electron in an orbit is $\propto \frac{Z^3}{n^3}$.

(D) Coulombic force of attraction on the electron is $\frac{Z^3}{n^4}$.

Choose the most appropriate answer from the options given below

(A) (A) only

(B) (C) only

(C) (A), (C) and (D) only

(D) (A) and (D) only

Solution:

$$(i) KE = -TE = 13.6 \times \frac{Z^2}{n^2} eV$$

$$\text{So, } KE \propto \frac{Z^2}{n^2}$$

$$(ii) V = 2.188 \times 10^6 \times \frac{Z}{n} \text{ m/sec}$$

$$\text{So, } Vn \propto Z$$

$$(iii) \text{Frequency} = \frac{v}{2\pi r}$$

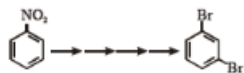
$$\text{So, } (F) \propto \frac{Z^2}{n^3}$$

$$(iv) \text{Force (F)} = \frac{kq_1q_2}{r^2}$$

$$\text{So, } F \propto \left(\frac{Z^3}{n^4} \right)$$

Q34:

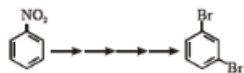
What is the correct sequence of reagents used for converting nitrobenzene into m-dibromobenzene ?



- (A) $\xrightarrow{\text{NaNO}_2} / \xrightarrow{\text{HCl}} / \xrightarrow{\text{KBr}} / \xrightarrow{\text{H}^+}$
- (B) $\xrightarrow{\text{Br}_2/\text{Fe}} / \xrightarrow{\text{Sn/HCl}} / \xrightarrow{\text{NaNO}_2/\text{HCl}} / \xrightarrow{\text{CuBr/HBr}}$
- (C) $\xrightarrow{\text{Sn/HCl}} / \xrightarrow{\text{KBr}} / \xrightarrow{\text{Br}_2} / \xrightarrow{\text{H}^+}$
- (D) $\xrightarrow{\text{Sn/HCl}} / \xrightarrow{\text{Br}_2} / \xrightarrow{\text{NaNO}_2} / \xrightarrow{\text{NaBr}}$

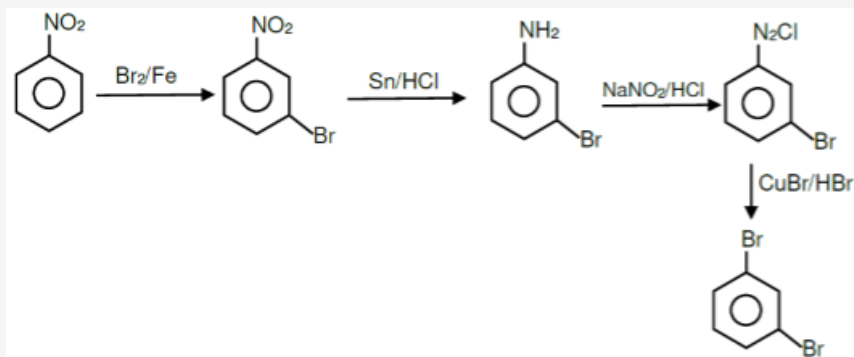
Q34:

What is the correct sequence of reagents used for converting nitrobenzene into m-dibromobenzene ?



- (A) $\xrightarrow{NaNO_2} / \xrightarrow{HCl} / \xrightarrow{KBr} / \xrightarrow{H^+}$
- (B) $\xrightarrow{Br_2/Fe} / \xrightarrow{Sn/HCl} / \xrightarrow{NaNO_2/HCl} / \xrightarrow{CuBr/HBr}$
- (C) $\xrightarrow{Sn/HCl} / \xrightarrow{KBr} / \xrightarrow{Br_2} / \xrightarrow{H^+}$
- (D) $\xrightarrow{Sn/HCl} / \xrightarrow{Br_2} / \xrightarrow{NaNO_2} / \xrightarrow{NaBr}$

Solution:



Q35: Most suitable salt which can be used for efficient clotting of blood will be

- (A) FeSO_4
- (B) NaHCO_3
- (C) $\text{Mg}(\text{HCO}_3)_2$
- (D) FeCl_3

Q35: Most suitable salt which can be used for efficient clotting of blood will be

- (A) FeSO_4
- (B) NaHCO_3
- (C) $\text{Mg}(\text{HCO}_3)_2$
- (D) FeCl_3

Solution:

Blood is a negative sol. According to Hardy-Schulz's rule the cation with high charge has high coagulation power. Hence ***FeCl_3*** can be used for clotting blood.

Q36: Match List - I and List - II.

List - I	List-II
(a) Valium	(i) Antifertility drug
(b) Morphine	(ii) Pernicious anaemia
(c) Norethindrone	(iii) Analgesic
(d) Vitamin B ₁₂	(iv) Tranquilizer

- (A) (a) – (i); (b) – (iii); (c) – (iv); (d) – (ii)
- (B) (a) – (iv); (b) - (iii); (c) – (ii); (d) – (iv)
- (C) (a) – (iv); (b) - (iii); (c) – (i); (d) – (ii)
- (D) (a) – (ii); (b) – (iv); (c) – (iii); (d) – (i)

Q36: Match List - I and List - II.

List - I	List-II
(a) Valium	(i) Antifertility drug
(b) Morphine	(ii) Pernicious anaemia
(c) Norethindrone	(iii) Analgesic
(d) Vitamin B ₁₂	(iv) Tranquilizer

(A) (a) – (i); (b) – (iii); (c) – (iv); (d) – (ii)

(B) (a) – (iv); (b) - (iii); (c) – (ii); (d) – (iv)

(C) (a) – (iv); (b) - (iii); (c) – (i); (d) – (ii)

(D) (a) – (ii); (b) – (iv); (c) – (iii); (d) – (i)

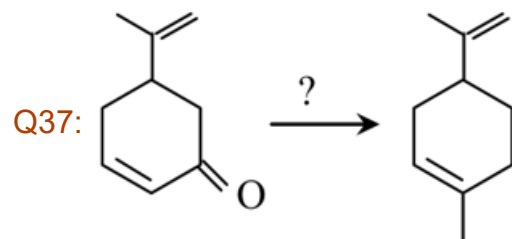
Solution:

Valium – Tranquilizer

Morphine – Analgesic

Norethindrone - Antifertility drug

Vitamin B₁₂ - Pernicious anaemia



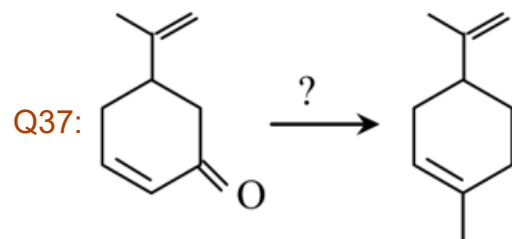
Which of the following reagent is suitable for the preparation of the product in the above reaction ?

(A) Red P + Cl₂

(B) NaBH₄

(C) NH₂-NH₂/C₂H₅[⊖]₃N[⊕]A

(D) Ni/H₂



Which of the following reagent is suitable for the preparation of the product in the above reaction ?

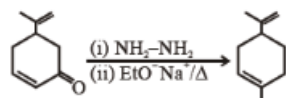
(A) Red P + Cl₂

(B) NaBH₄

(C) NH₂-NH₂/C₂H₅[⊖]ON[⊕]A

(D) Ni/H₂

Solution:



To reduce the carbonyl groups into alkane wolff kischner reduction is used, without affecting the double bond.

Q38: In polymer Buna-S : 'S' stands for :

- (A) Strength
- (B) Sulphur
- (C) Sulphonation
- (D) Styrene

Q38: In polymer Buna-S : 'S' stands for :

- (A) Strength
- (B) Sulphur
- (C) Sulphonation
- (D) Styrene

Solution:

Buna – S is an addition polymer of 1, 3 – Butadiene and styrene.

So, 'S' stands for styrene ($Ph - CH = CH_2$)

Q39: Given below are two statements:

Statement I: The value of the parameter "Biochemical Oxygen Demand (BOD)" is important for survival of aquatic life.

Statement II : The optimum value of BOD is 6.5 ppm.

In the light of the above statements, choose the most appropriate answer from the options given below

- (A) Statement I is true but Statement II is false
- (B) Both Statement I and Statement II are false
- (C) Statement I is false but Statement II is true
- (D) Both Statement I and Statement II are true

Q39: Given below are two statements:

Statement I: The value of the parameter "Biochemical Oxygen Demand (BOD)" is important for survival of aquatic life.

Statement II : The optimum value of BOD is 6.5 ppm.

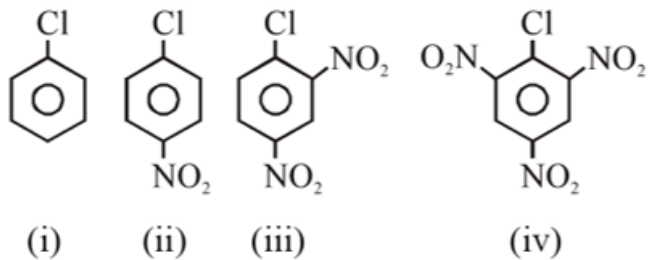
In the light of the above statements, choose the most appropriate answer from the options given below

- (A) Statement I is true but Statement II is false
- (B) Both Statement I and Statement II are false
- (C) Statement I is false but Statement II is true
- (D) Both Statement I and Statement II are true

Solution:

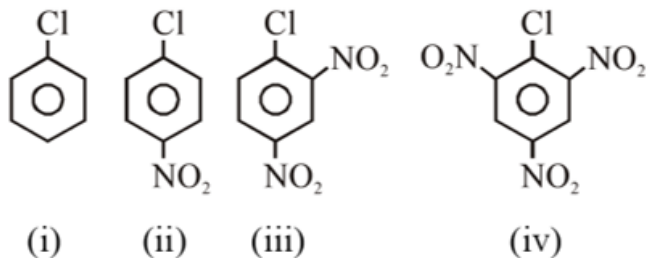
Clean water would have BOD value of less than 5 ppm whereas highly polluted water could have a BOD value of 17 ppm or more.

Q40: The correct order of the following compounds showing increasing tendency towards nucleophilic substitution reaction is



- (A) (i) < (ii) < (iii) < (iv)
- (B) (iv) < (i) < (ii) < (iii)
- (C) (iv) < (iii) < (ii) < (i)
- (D) (iv) < (i) < (iii) < (ii)

Q40: The correct order of the following compounds showing increasing tendency towards nucleophilic substitution reaction is



(A) (i) < (ii) < (iii) < (iv)

(B) (iv) < (i) < (ii) < (iii)

(C) (iv) < (iii) < (ii) < (i)

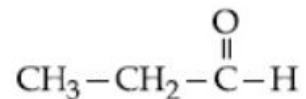
(D) (iv) < (i) < (iii) < (ii)

Solution:

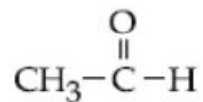
Electron withdrawing group increases rate of aromatic nucleophilic substitution reaction.

Q41: Which one of the following carbonyl compounds cannot be prepared by addition of water on an alkyne in the presence of HgSO_4 and H_2SO_4 ?

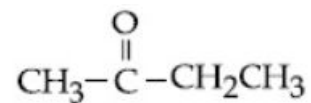
(A)



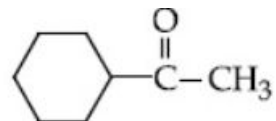
(B)



(C)

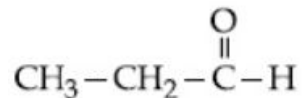


(D)

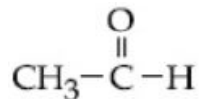


Q41: Which one of the following carbonyl compounds cannot be prepared by addition of water on an alkyne in the presence of HgSO_4 and H_2SO_4 ?

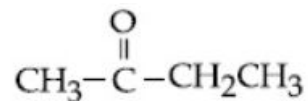
(A)



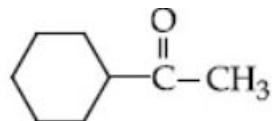
(B)



(C)



(D)



Solution:

$\text{HgSO}_4/\text{H}_2\text{SO}_4/\text{H}_2\text{O}$ gives Markovnikov's addition product with any alkyne which will give ketone as a product. Hence $\text{CH}_3-\text{CH}_2-\text{CHO}$ will not be produced.

Q42: The incorrect statement among the following is:

- (A) Red colour of ruby is due to the presence of Co^{3+}
- (B) If the t_{2g} and e_g orbitals of an octahedral compound are completely filled, it is predicted to be colourless.
- (C) Secondary valency of a coordination complex is equal to the coordination number
- (D) Removal of water from $[\text{Ti}(\text{H}_2\text{O})_6]\text{Cl}_3$ on heating renders it colourless.

Q42: The incorrect statement among the following is:

- (A) Red colour of ruby is due to the presence of Co^{3+}
- (B) If the t_{2g} and e_g orbitals of an octahedral compound are completely filled, it is predicted to be colourless.
- (C) Secondary valency of a coordination complex is equal to the coordination number
- (D) Removal of water from $[\text{Ti}(\text{H}_2\text{O})_6]\text{Cl}_3$ on heating renders it colourless.

Solution:

Red colour of Ruby is due to Cr^{3+} not due to Co^{3+}

Q43: The correct set from the following in which both pairs are in correct order of melting point is

- (A) $\text{LiF} > \text{LiCl}$; $\text{NaCl} > \text{MgO}$
- (B) $\text{LiCl} > \text{LiF}$; $\text{NaCl} > \text{MgO}$
- (C) $\text{LiCl} > \text{LiF}$; $\text{MgO} > \text{NaCl}$
- (D) $\text{LiF} > \text{LiCl}$; $\text{MgO} > \text{NaCl}$

Q43: The correct set from the following in which both pairs are in correct order of melting point is

- (A) $\text{LiF} > \text{LiCl}$; $\text{NaCl} > \text{MgO}$
- (B) $\text{LiCl} > \text{LiF}$; $\text{NaCl} > \text{MgO}$
- (C) $\text{LiCl} > \text{LiF}$; $\text{MgO} > \text{NaCl}$
- (D) $\text{LiF} > \text{LiCl}$; $\text{MgO} > \text{NaCl}$

Solution:

$$\text{Lattice energy} \propto \frac{|Z^+| |Z^-|}{r_+ + r_-}$$

Lattice energy $\propto$ melting point

Lattice energy : $\text{LiF} > \text{LiCl}$

Lattice energy : $\text{MgO} > \text{NaCl}$

M.Pt. of $\text{LiF} = 845^\circ\text{C}$

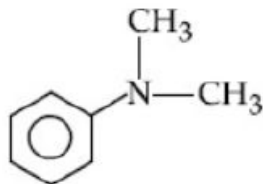
M.Pt. of $\text{LiCl} = 605^\circ\text{C}$

M.Pt. of $\text{NaCl} = 801^\circ\text{C}$

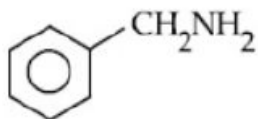
M.Pt. of $\text{MgO} = 2852^\circ\text{C}$

Q44: The diazonium salt of which of the following compounds will form a coloured dye on reaction with β -Naphthol in NaOH ?

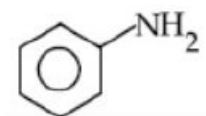
(A)



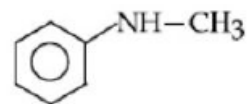
(B)



(C)

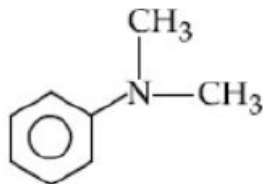


(D)

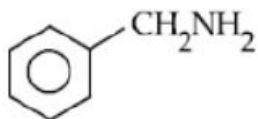


Q44: The diazonium salt of which of the following compounds will form a coloured dye on reaction with β -Naphthol in NaOH ?

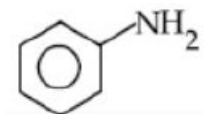
(A)



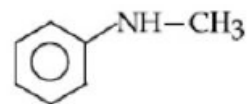
(B)



(C)



(D)



Solution:

Only aromatic primary amines will give Azo dye test.

Q45: Match List – I and List – II.

List I	List II
(a) $\text{R}-\overset{\text{O}}{\parallel}{\text{C}}-\text{Cl} \rightarrow \text{R}-\text{CHO}$	(i) Br_2/NaOH
(b) $\text{R}-\text{CH}_2-\text{COOH} \rightarrow \text{R}-\underset{\text{Cl}}{\text{CH}}-\text{COOH}$	(ii) $\text{H}_2/\text{Pd}-\text{BaSO}_4$
(c) $\text{R}-\overset{\text{O}}{\parallel}{\text{C}}-\text{NH}_2 \rightarrow \text{R}-\text{NH}_2$	(iii) $\text{Zn}(\text{Hg})/\text{Conc. HCl}$
(d) $\text{R}-\overset{\text{O}}{\parallel}{\text{C}}-\text{CH}_3 \rightarrow \text{R}-\text{CH}_2-\text{CH}_3$	(iv) $\text{Cl}_2/\text{Red P, H}_2\text{O}$

Choose the correct answer from the options given below:

- (A) (a)-(iii), (b)-(iv), (c)-(i), (d)-(ii)
- (B) (a)-(ii), (b)-(iv), (c)-(i), (d)-(iii)
- (C) (a)-(iii), (b)-(i), (c)-(iv), (d)-(ii)
- (D) (a)-(ii), (b)-(i), (c)-(iv), (d)-(iii)

Q45: Match List – I and List – II.

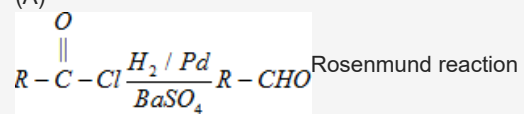
List I	List II
(a) $\text{R}-\overset{\text{O}}{\parallel}{\text{C}}-\text{Cl} \rightarrow \text{R}-\text{CHO}$	(i) Br_2/NaOH
(b) $\text{R}-\text{CH}_2-\text{COOH} \rightarrow \text{R}-\underset{\text{Cl}}{\text{CH}}-\text{COOH}$	(ii) $\text{H}_2/\text{Pd}-\text{BaSO}_4$
(c) $\text{R}-\overset{\text{O}}{\parallel}{\text{C}}-\text{NH}_2 \rightarrow \text{R}-\text{NH}_2$	(iii) $\text{Zn}(\text{Hg})/\text{Conc. HCl}$
(d) $\text{R}-\overset{\text{O}}{\parallel}{\text{C}}-\text{CH}_3 \rightarrow \text{R}-\text{CH}_2-\text{CH}_3$	(iv) $\text{Cl}_2/\text{Red P, H}_2\text{O}$

Choose the correct answer from the options given below:

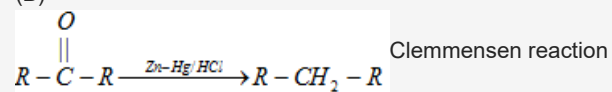
- (A) (a)-(iii), (b)-(iv), (c)-(i), (d)-(ii)
- (B) (a)-(ii), (b)-(iv), (c)-(i), (d)-(iii)
- (C) (a)-(iii), (b)-(i), (c)-(iv), (d)-(ii)
- (D) (a)-(ii), (b)-(i), (c)-(iv), (d)-(iii)

Solution:

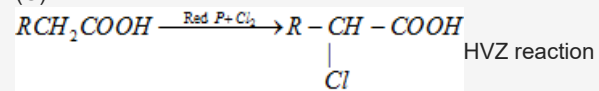
(A)



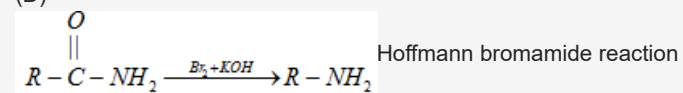
(B)



(C)



(D)



Q46: Match List - I with List - II

List – I (Salt)	List - II (Flame colour wavelength)
(a) LiCl	(i) 455.5 nm
(b) NaCl	(ii) 670.8 nm
(c) RbCl	(iii) 780.0 nm
(d) CsCl	(iv) 589.2 nm

Choose the correct answer from the options given below :

- (A) (a)-(ii), (b)-(iv), (c)-(iii), (d)-(i)
- (B) (a)-(iv), (d)-(ii), (c)-(iii), (d)-(i)
- (C) (a)-(ii), (b)-(i), (c)-(iv), (d)-(iii)
- (D) (a)-(i), (b)-(iv), (c)-(ii), (d)-(iii)

Q46: Match List - I with List - II

List – I (Salt)	List - II (Flame colour wavelength)
(a) LiCl	(i) 455.5 nm
(b) NaCl	(ii) 670.8 nm
(c) RbCl	(iii) 780.0 nm
(d) CsCl	(iv) 589.2 nm

Choose the correct answer from the options given below :

- (A) (a)-(ii), (b)-(iv), (c)-(iii), (d)-(i)
- (B) (a)-(iv), (d)-(ii), (c)-(iii), (d)-(i)
- (C) (a)-(ii), (b)-(i), (c)-(iv), (d)-(iii)
- (D) (a)-(i), (b)-(iv), (c)-(ii), (d)-(iii)

Solution:

The alkali metals and their salts impart characteristic colour to an oxidizing flame. This is because the heat from the flame excites the outermost orbital electron to a higher energy level. When the excited electron comes back to the ground state, there is emission of radiation in the visible region as given below

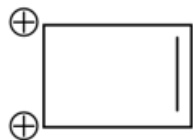
Metal	Li	Na	K	Rb	Cs
Colour	Crimson red	Yellow	Violet	Red violet	Blue
λ/nm	670.8	589.2	766.5	780.0	455.5

Q47: Which one of the following compounds is non-aromatic?

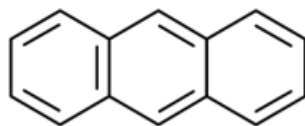
(A)



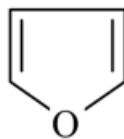
(B)



(C)



(D)

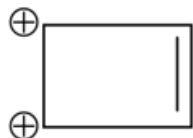


Q47: Which one of the following compounds is non-aromatic?

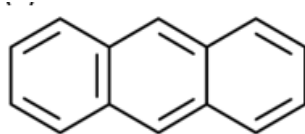
(A)



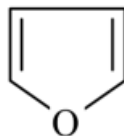
(B)



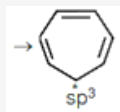
(C)



(D)



Solution:



contains sp^3 — C atom. So it is non-planar & hence non aromatic.

→ All other options follow Huckel's rule i.e. $(4n + 2) \pi$ electrons. So all are aromatic compounds.

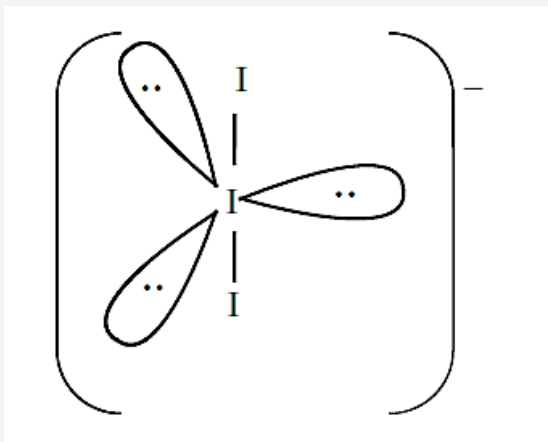
Q48: The correct shape and I-I-I bond angles respectively in I_3^- ion are :

- (A) Distorted trigonal planar 135° and 90°
- (B) T-shaped; 180° and 90°
- (C) Trigonal planar; 120°
- (D) Linear; 180°

Q48: The correct shape and I-I-I bond angles respectively in I_3^- ion are :

- (A) Distorted trigonal planar 135° and 90°
- (B) T-shaped; 180° and 90°
- (C) Trigonal planar; 120°
- (D) Linear; 180°

Solution:



Q49: The calculated magnetic moments (spin only value) for species

$[FeCl_4]^{2-}$, $[Co(C_2O_4)_3]^{3-}$ and MnO_4^{2-} respectively:

- (A) 4.90, 0 and 1.73 BM
- (B) 5.82, 0 and 0 BM
- (C) 5.92, 4.90 and 0 BM
- (D) 4.90, 0 and 2.83 BM

Q49: The calculated magnetic moments (spin only value) for species $[FeCl_4]^{2-}$, $[Co(C_2O_4)_3]^{3-}$ and MnO_4^{2-} respectively:

- (A) 4.90, 0 and 1.73 BM
- (B) 5.82, 0 and 0 BM
- (C) 5.92, 4.90 and 0 BM
- (D) 4.90, 0 and 2.83 BM

Solution:

Complex

- (i) $[FeCl_4]^{2-} = Fe^{2+} = 3d^6$, $eg^{2,1}t_2g^{1,1,1}$, $\mu = 4.9 \text{ BM}$
- (ii) $[Co(C_2O_4)_3]^{3-} = Co^{3+} = 3d^6$, $t_2g^{2,2,2}$, $eg^{0,0}$, $\mu = 0 \text{ BM}$
- (iii) $MnO_4^{2-} = Mn^{+6} = 3d^1$, $eg^{1,0}t_2g^{0,0,0}$, $\mu = 1.73 \text{ BM}$

Q50:

Which is the correct order of the following elements with respect to their density?

(A) $\text{Cr} < \text{Zn} < \text{Co} < \text{Cu} < \text{Fe}$

(B) $\text{Zn} < \text{Cu} < \text{Co} < \text{Fe} < \text{Cr}$

(C) $\text{Zn} < \text{Cr} < \text{Fe} < \text{Co} < \text{Cu}$

(D) $\text{Cr} < \text{Fe} < \text{Co} < \text{Cu} < \text{Zn}$

Q50:

Which is the correct order of the following elements with respect to their density?

(A) $\text{Cr} < \text{Zn} < \text{Co} < \text{Cu} < \text{Fe}$

(B) $\text{Zn} < \text{Cu} < \text{Co} < \text{Fe} < \text{Cr}$

(C) $\text{Zn} < \text{Cr} < \text{Fe} < \text{Co} < \text{Cu}$

(D) $\text{Cr} < \text{Fe} < \text{Co} < \text{Cu} < \text{Zn}$

Solution:

According to NCERT data, the densities (in g cm^{-3}) are:

$\text{Zn} = 7.1$

$\text{Cr} = 7.19$

$\text{Fe} = 7.8$

$\text{Co} = 8.7$

$\text{Cu} = 8.9$

Q51: Among the following allotropic forms of sulphur, the number of allotropic forms, which will show paramagnetism is

(A) α -sulphur

(B) β -sulphur

(C) S_2 -form

Q51: Among the following allotropic forms of sulphur, the number of allotropic forms, which will show paramagnetism is

(A) α -sulphur

(B) β -sulphur

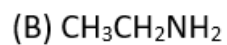
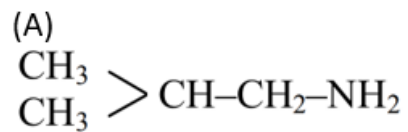
(C) S_2 -form

1.00

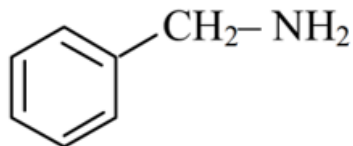
Solution:

In S_2 , molecules have 2 unpaired electrons similar to O_2 according to MOT.

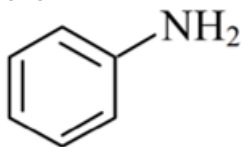
Q52: The total number of amines among the following which can be synthesized by Gabriel synthesis is



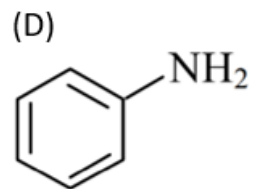
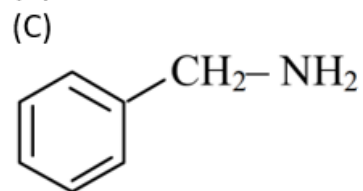
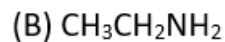
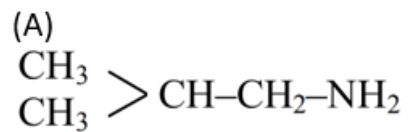
(C)



(D)



Q52: The total number of amines among the following which can be synthesized by Gabriel synthesis is



3.00

Solution:

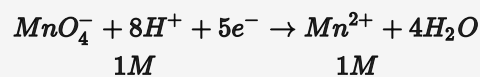
Only primary aliphatic amines can be formed by Gabriel-phthalimide reaction.

Q53: The magnitude of the change in oxidising power of the MnO_4^- / Mn^{2+} couple is $x \times 10^{-4}$ V, if the H^+ concentration is decreased from 1 M to 10^{-4} M at $25^\circ C$. (Assume concentration of MnO_4^- and Mn^{2+} to be same on change in H^+ concentration). The value of x is (Rounded off to the nearest integer) [Given : $\frac{2.303RT}{F} = 0.059$]

Q53: The magnitude of the change in oxidising power of the MnO_4^- / Mn^{2+} couple is $x \times 10^{-4}$ V, if the H^+ concentration is decreased from 1 M to 10^{-4} M at 25°C . (Assume concentration of MnO_4^- and Mn^{2+} to be same on change in H^+ concentration). The value of x is (Rounded off to the nearest integer) [Given : $\frac{2.303RT}{F} = 0.059$]

3776.00

Solution:



$$E = E^\circ - \frac{0.059}{5} \log \frac{[Mn^{2+}]}{[MnO_4^-] [H^+]^8}$$

$$E_1 = E^\circ - \frac{0.059}{5} \log 1 = E^\circ$$

$$E_2 = E^\circ - \frac{0.059}{5} \log \frac{1}{(10^{-4})^8}$$

$$= E^\circ - \frac{0.059}{5} \log 10^{32}$$

$$= E^\circ - \frac{0.059}{5} \times 32$$

So, magnitude of change in electrode potential

$$|(E_2 - E_1)| = \left| -\frac{0.059 \times 32}{5} \right| = 0.3776 = 3776 \times 10^{-4}$$

Q54: The volume occupied by 4.75 g of acetylene gas at 50 °C and 740 mmHg pressure is _____ L (Rounded off to the nearest integer) [Given $R = 0.082 \text{ L atm K}^{-1} \text{ mol}^{-1}$]

Q54: The volume occupied by 4.75 g of acetylene gas at 50 °C and 740 mmHg pressure is _____ L (Rounded off to the nearest integer) [Given R = 0.082 L atm K⁻¹ mol⁻¹]

5.00

Solution:

$$PV = nRT$$

$$\frac{740}{760} \times V = \frac{4.75}{26} \times 0.082 \times 323$$

$$V = 4.97 \text{ lit}$$

So Ans. = 5

Q55: The formula of a gaseous hydrocarbon which requires 6 times of its own volume of O_2 for complete oxidation and produces 4 times its own volume of CO_2 is C_xH_y . The value of y is

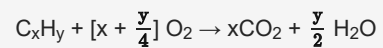
_____.

Q55: The formula of a gaseous hydrocarbon which requires 6 times of its own volume of O_2 for complete oxidation and produces 4 times its own volume of CO_2 is C_xH_y . The value of y is

8.00

Solution:

The balanced reaction is as follows:



Let, a litre of C_xH_y be used

$$a[x + \frac{y}{4}] = 6a \dots (1)$$

$$ax = 4a \dots (2)$$

$$\therefore x = 4 \text{ and } y = 8$$

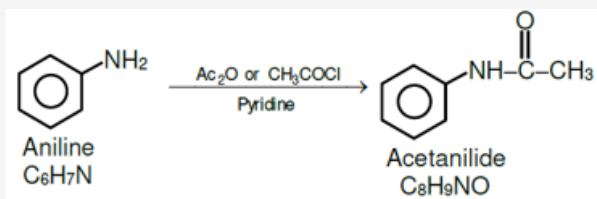
Hence, the gas is C_4H_8

Q56: 1.86 g of aniline completely reacts to form acetanilide. 10% of the product is lost during purification. Amount of acetanilide obtained after purification (in g) is ____ $\times 10^{-2}$.

Q56: 1.86 g of aniline completely reacts to form acetanilide. 10% of the product is lost during purification. Amount of acetanilide obtained after purification (in g) is ____ $\times 10^{-2}$.

243.00

Solution:



Molar mass = 93 Molar mass = 135

93 g aniline produce 135 g acetanilide

1.86 g aniline produces $\frac{135 \times 1.86}{93} = 2.70\text{g}$

At 90% yield of reaction it produces $= \frac{2.70 \times 90}{100} = 2.43\text{g} \approx 243 \times 10^{-2}\text{g}$

Q57: Sucrose hydrolyses in acid solution into glucose and fructose following first order rate law with a half-life of 3.33 h at 25 °C. After 9 h, the fraction of sucrose remaining is f . The value of $\log_{10} \left(\frac{1}{f} \right)$ is _____ $\times 10^{-2}$. (Rounded off to nearest integer) [Assume: $\ln 10 = 2.303$, $\ln 2 = 0.693$]

Q57: Sucrose hydrolyses in acid solution into glucose and fructose following first order rate law with a half-life of 3.33 h at 25 °C. After 9 h, the fraction of sucrose remaining is f . The value of $\log_{10} \left(\frac{1}{f} \right)$ is _____ $\times 10^{-2}$. (Rounded off to nearest integer) [Assume: $\ln 10 = 2.303$, $\ln 2 = 0.693$]

81.00

Solution:

$$\begin{aligned}
 n &= \frac{\text{time given}}{(T_{1/2})} = \frac{9}{3.33} = 2.7 \\
 C_n &= \frac{C_0}{(2)^n} \text{ \& } F = \left(\frac{C_n}{C_0} \right) = \frac{1}{(2)^n} = \frac{1}{(2)^{2.7}} \\
 \log \left(\frac{1}{F} \right) &= \log (2)^{2.7} \\
 &= 2.7 \times \log 2 \\
 &= 0.81 \approx 81 \times 10^{-2}
 \end{aligned}$$

Q58: The solubility product of PbI_2 is 8.0×10^{-9} . The solubility of lead iodide in 0.1 molar solution of lead nitrate is $x \times 10^{-6}$ mol/L. The value of x is _____. (Rounded off to the nearest integer)

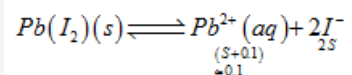
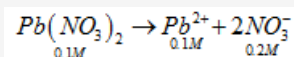
[Given $\sqrt{2} = 1.41$]

Q58: The solubility product of PbI_2 is 8.0×10^{-9} . The solubility of lead iodide in 0.1 molar solution of lead nitrate is $x \times 10^{-6}$ mol/L. The value of x is _____. (Rounded off to the nearest integer)

[Given $\sqrt{2} = 1.41$]

141.00

Solution:



$$K_{sp}(PbI_2) = [Pb^{2+}][I^-]^2 = 8 \times 10^{-9}$$

$$= 0.1(2S)^2 = 8 \times 10^{-9}$$

$$= 4S^2 = 8 \times 10^{-9}$$

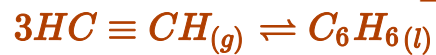
$$S^2 = 2 \times 10^{-9}$$

$$S = \sqrt{2} \times 10^{-4}$$

$$S = 1.414 \times 10^{-4} \approx 141.4 \times 10^{-6}$$

Q59: Assuming ideal behaviour, the magnitude of log K for the following reaction at 25° C is

$x \times 10^{-1}$. The value of x is ____ (Integer answer)



[Given:

$$\Delta_f G^\circ (HC \equiv CH) = -2.04 \times 10^5 \text{ J mol}^{-1}; \Delta_f G^\circ (C_6H_6) = -1.24 \times 10^5 \text{ J mol}^{-1}; R = 8.314 \text{ J K}^{-1} \text{ mol}^{-1}]$$

Q59: Assuming ideal behaviour, the magnitude of $\log K$ for the following reaction at 25°C is $x \times 10^{-1}$. The value of x is ____ (Integer answer)



[Given:

$$\Delta_f G^\circ (HC \equiv CH) = -2.04 \times 10^5 \text{ J mol}^{-1}; \Delta_f G^\circ (C_6H_6) = -1.24 \times 10^5 \text{ J mol}^{-1}; R = 8.314 \text{ J K}^{-1} \text{ mol}^{-1}]$$

855.00

Solution:

$$\Delta G_{rxn}^\circ = \Delta G_f^\circ (C_6H_6) - 3\Delta G_f^\circ (C_2H_2)$$

$$\Delta G_{rxn}^\circ = -1.24 \times 10^5 - 3 \times -(2.04 \times 10^5)$$

$$\Delta G_{rxn}^\circ = 4.88 \times 10^5 \text{ J/mol}$$

$$-2.303 RT \log K = 4.88 \times 10^5 \text{ J}$$

$$|\log K| = |-85.5| = |-855 \times 10^{-1}|$$

$$= 855 \times 10^{-1}$$

So Ans. 855

Q60: C_6H_6 freezes at 5.5°C . The temperature at which a solution of 10 g of C_4H_{10} in 200 g of C_6H_6 freeze is ____ $^\circ\text{C}$. (The molal freezing point depression constant of C_6H_6 , is 5.12°C/m)

Q60: C_6H_6 freezes at 5.5°C . The temperature at which a solution of 10 g of C_4H_{10} in 200 g of C_6H_6 freeze is ____ $^\circ\text{C}$. (The molal freezing point depression constant of C_6H_6 , is 5.12°C/m)

1.00

Solution:

$$\begin{aligned}\Delta T_f &= k_f \times m \\ &= 5.12 \times \left(\frac{10 \times 1000}{58 \times 200} \right) \\ T_f - (T_f)_{\text{soln}} &= 4.41 \\ 5.5 - (T_f)_{\text{soln}} &= 4.41 \\ (T_f)_{\text{soln}} &= 1.09^\circ\text{C}\end{aligned}$$

Q61: Let $f(x)$ be a differentiable function defined on $[0,2]$ such that $f'(x) = f'(2-x)$ for all $x \in (0,2)$, $f(0) = 1$ and $f(2) = e^2$. Then the value of $\int_0^2 f(x) dx$ is

(A) $2(1 - e^2)$

(B) $1 - e^2$

(C) $2(1 + e^2)$

(D) $1 + e^2$

Q61: Let $f(x)$ be a differentiable function defined on $[0,2]$ such that $f'(x) = f'(2-x)$ for all $x \in (0,2)$, $f(0) = 1$ and $f(2) = e^2$. Then the value of $\int_0^2 f(x) dx$ is

(A) $2(1 - e^2)$

(B) $1 - e^2$

(C) $2(1 + e^2)$

(D) $1 + e^2$

Solution:

$$f'(x) = f'(2-x)$$

On integrating both side $f(x) = -f(2-x) + c$

Put $x = 0$

$$f(0) + f(2) = c \Rightarrow c = 1 + e^2$$

$$\Rightarrow f(x) + f(2-x) = 1 + e^2 \quad \dots (i)$$

$$I = \int_0^2 f(x) dx = \int_0^1 \{f(x) + f(2-x)\} dx = (1 + e^2)$$

Q62: A possible value of $\tan\left(\frac{1}{4}\sin^{-1}\frac{\sqrt{63}}{8}\right)$ is :

(A) $2\sqrt{2} - 1$

(B) $\frac{1}{\sqrt{7}}$

(C) $\frac{1}{2\sqrt{2}}$

(D) $\sqrt{7} - 1$

Q62: A possible value of $\tan\left(\frac{1}{4}\sin^{-1}\frac{\sqrt{63}}{8}\right)$ is :

(A) $2\sqrt{2} - 1$

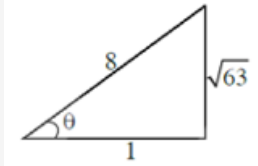
(B) $\frac{1}{\sqrt{7}}$

(C) $\frac{1}{2\sqrt{2}}$

(D) $\sqrt{7} - 1$

Solution:

Let $\sin^{-1}\frac{\sqrt{63}}{8} = \theta \Rightarrow \sin \theta = \frac{\sqrt{63}}{8}$



$$\tan\left(\frac{1}{4}\sin^{-1}\frac{\sqrt{63}}{8}\right) = \tan\left(\frac{\theta}{4}\right) = \frac{1 - \cos \frac{\theta}{2}}{\sin \frac{\theta}{2}} = \frac{1 - \sqrt{\frac{1 + \cos \theta}{2}}}{\sqrt{\frac{1 - \cos \theta}{2}}} = \frac{1 - \frac{3}{4}}{\frac{\sqrt{7}}{4}} = \frac{1}{\sqrt{7}}$$

Q63: For which of the following curves, the line $x + \sqrt{3}y = 2\sqrt{3}$ is the tangent at the point $\left(\frac{3\sqrt{3}}{2}, \frac{1}{2}\right)$?

(A) $2x^2 - 18y^2 = 9$

(B) $x^2 + 9y^2 = 9$

(C) $y^2 = \frac{1}{6\sqrt{3}}x$

(D) $x^2 + y^2 = 7$

Q63: For which of the following curves, the line $x + \sqrt{3}y = 2\sqrt{3}$ is the tangent at the point $\left(\frac{3\sqrt{3}}{2}, \frac{1}{2}\right)$?

(A) $2x^2 - 18y^2 = 9$

(B) $x^2 + 9y^2 = 9$

(C) $y^2 = \frac{1}{6\sqrt{3}}x$

(D) $x^2 + y^2 = 7$

Solution:

Tangent to $x^2 + 9y^2 = 9$ at point $\left(\frac{3\sqrt{3}}{2}, \frac{1}{2}\right)$ is $x\left(\frac{3\sqrt{3}}{2}\right) + 9y\left(\frac{1}{2}\right) = 9$

Q64: The value of the integral $\int_1^3 [x^2 - 2x - 2] dx$, where $[x]$ denotes the greatest integer less than or equal to x , is

- (A) -5
- (B) $-\sqrt{2} - \sqrt{3} - 1$
- (C) -4
- (D) $-\sqrt{2} - \sqrt{3} + 1$

Q64: The value of the integral $\int_1^3 [x^2 - 2x - 2] dx$, where $[x]$ denotes the greatest integer less than or equal to x , is

(A) -5

(B) $-\sqrt{2} - \sqrt{3} - 1$

(C) -4

(D) $-\sqrt{2} - \sqrt{3} + 1$

Solution:

$$I = \int_1^3 -3 dx + \int_1^3 [(x-1)^2] dx \quad x-1 = t; dx = dt$$

$$I = (-6) + \int_0^2 [t^2] dt$$

$$I = -6 + \int_0^1 0 dt + \int_1^{\sqrt{2}} 1 dt + \int_{\sqrt{2}}^{\sqrt{3}} 2 dt + \int_{\sqrt{3}}^2 3 dt$$

$$I = (-6) + (\sqrt{2} - 1) + 2\sqrt{3} - 2\sqrt{2} + 6 - 3\sqrt{3}$$

$$I = -1 - \sqrt{2} - \sqrt{3}$$

Q65: Let a, b, c be in arithmetic progression. Let the centroid of the triangle with vertices $(a, c), (2, b)$ and (a, b) be $(\frac{10}{3}, \frac{7}{3})$. If α, β are the roots of the equation $ax^2 + bx + 1 = 0$, the value of $\alpha^2 + \beta^2 - \alpha\beta$ is

(A) $-\frac{71}{256}$

(B) $\frac{69}{256}$

(C) $-\frac{69}{256}$

(D) $\frac{71}{256}$

Q65: Let a, b, c be in arithmetic progression. Let the centroid of the triangle with vertices $(a, c), (2, b)$ and (a, b) be $(\frac{10}{3}, \frac{7}{3})$. If α, β are the roots of the equation $ax^2 + bx + 1 = 0$, the value of $\alpha^2 + \beta^2 - \alpha\beta$ is

(A) $-\frac{71}{256}$

(B) $\frac{69}{256}$

(C) $-\frac{69}{256}$

(D) $\frac{71}{256}$

Solution:

$$2b = a + c$$

$$\frac{2a+2}{3} = \frac{10}{3} \text{ and } \frac{2b+c}{3} = \frac{7}{3}$$

$$\Rightarrow \left. \begin{array}{l} a = 4 \\ 2b + c = 7 \\ 2b - c = 4 \end{array} \right\} \text{ solving,}$$

$$b = \frac{11}{4} \quad c = \frac{3}{2}$$

$$\therefore \text{ Quadratic Equation is } 4x^2 + \frac{11}{4}x + 1 = 0$$

$$\therefore \text{ The value of } (\alpha + \beta)^2 - 3\alpha\beta = \frac{121}{256} - \frac{3}{4} = -\frac{71}{256}$$

Q66: Let f be a twice differentiable function defined on $\mathbb{R}$ such that $f(0) = 1$, $f'(0) = 2$ and

$f'(x) \neq 0$ for all $x \in \mathbb{R}$. If $\begin{vmatrix} f(x) & f'(x) \\ f'(x) & f''(x) \end{vmatrix} = 0$, for all $x \in \mathbb{R}$, then the value of $f(1)$

lies in the interval

- (A) $(0,3)$
- (B) $(9,12)$
- (C) $(3,6)$
- (D) $(6,9)$

Q66: Let f be a twice differentiable function defined on $\mathbb{R}$ such that $f(0) = 1$, $f'(0) = 2$ and $f'(x) \neq 0$ for all $x \in \mathbb{R}$. If $\left| \begin{matrix} f(x) & f'(x) \\ f'(x) & f''(x) \end{matrix} \right| = 0$, for all $x \in \mathbb{R}$, then the value of $f(1)$ lies in the interval

- (A) (0,3)
- (B) (9,12)
- (C) (3,6)
- (D) (6,9)

Solution:

$$\text{Given } f(x) f''(x) - (f'(x))^2 = 0$$

$$\text{Let } h(x) = \frac{f(x)}{f'(x)}$$

$$\Rightarrow h'(x) = 0 \quad \Rightarrow h(x) = k$$

$$\Rightarrow \frac{f(x)}{f'(x)} = k \quad \Rightarrow f(x) = k f'(x)$$

$$\Rightarrow f(0) = k f'(0) \Rightarrow 1 = k(2) \Rightarrow k = \frac{1}{2}$$

$$\text{Now } f(x) = \frac{1}{2} f'(x) \Rightarrow \int 2 dx = \int \frac{f'(x)}{f(x)} dx$$

$$\Rightarrow 2x = \ln|f(x)| + C$$

$$\text{As } f(0) = 1 \Rightarrow C = 0$$

$$\Rightarrow 2x = \ln|f(x)| \Rightarrow f(x) = \pm e^{2x}$$

$$\text{As } f(x) = 1 \Rightarrow f(x) = e^{2x} \Rightarrow f(1) = e^2$$

Q67: For the system of linear equations:

$$x - 2y = 1, x - y + kz = -2, ky + 4z = 6, k \in R$$

consider the following statements :

(A) The system has unique solution if $k \neq 2, k \neq -2$.

(B) The system has unique solution if $k = -2$.

(C) The system has unique solution if $k = 2$.

(D) The system has no-solution if $k = 2$.

(E) The system has infinite number of solutions if $k \neq -2$.

Which of the following statements are correct ?

(A) (B) and (E) only

(B) (A) and (E) only

(C) (A) and (D) only

(D) (C) and (D) only

Q67: For the system of linear equations:

$$x - 2y = 1, x - y + kz = -2, ky + 4z = 6, k \in R$$

consider the following statements :

(A) The system has unique solution if $k \neq 2, k \neq -2$.

(B) The system has unique solution if $k = -2$.

(C) The system has unique solution if $k = 2$.

(D) The system has no-solution if $k = 2$.

(E) The system has infinite number of solutions if $k \neq -2$.

Which of the following statements are correct ?

(A) (B) and (E) only

(B) (A) and (E) only

(C) (A) and (D) only

(D) (C) and (D) only

Solution:

$$D = \begin{vmatrix} 1 & -2 & 0 \\ 1 & -1 & k \\ 0 & k & 4 \end{vmatrix} = 4 - k^2$$

$$D_1 = \begin{vmatrix} 1 & -2 & 0 \\ -2 & -1 & k \\ 6 & k & 4 \end{vmatrix} = -(k+10)(k+2)$$

For unique solution $D \neq 0 \Rightarrow k \neq \pm 2$

For no solution, $D = 0, D_1 \neq 0$

$\Rightarrow k = 2$

At $k = -2, D_1 = D_2 = D_3 = D = 0$

So system has infinite solution.

Q68: Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be defined as

$$f(x) = \begin{cases} -55x, & \text{if } x < -5 \\ 2x^3 - 3x^2 - 120x, & \text{if } -5 \leq x \leq 4 \\ 2x^3 - 3x^2 - 36x - 336, & \text{if } x > 4, \end{cases}$$

Let $A = \{x \in \mathbb{R} : f \text{ is increasing}\}$. Then A is equal to

- (A) $(-\infty, -5) \cup (4, \infty)$
- (B) $(-5, \infty)$
- (C) $(-5, -4) \cup (4, \infty)$
- (D) $(-\infty, -5) \cup (-4, \infty)$

Q68: Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be defined as

$$f(x) = \begin{cases} -55x, & \text{if } x < -5 \\ 2x^3 - 3x^2 - 120x, & \text{if } -5 \leq x \leq 4 \\ 2x^3 - 3x^2 - 36x - 336, & \text{if } x > 4, \end{cases}$$

Let $A = \{x \in \mathbb{R} : f \text{ is increasing}\}$. Then A is equal to

- (A) $(-\infty, -5) \cup (4, \infty)$
- (B) $(-5, \infty)$
- (C) $(-5, -4) \cup (4, \infty)$
- (D) $(-\infty, -5) \cup (-4, \infty)$

Solution:

$$f'(x) = \begin{cases} -55 & ; x < -5 \\ 6(x^2 - x - 20) & ; -5 < x < 4 \\ 6(x^2 - x - 6) & ; x > 4 \end{cases}$$

$$f'(x) = \begin{cases} -55 & ; x < -5 \\ 6(x-5)(x+4) & ; -5 < x < 4 \\ 6(x-3)(x+2) & ; x > 4 \end{cases}$$

Hence, $f(x)$ is monotonically increasing is $(-5, -4) \cup (4, \infty)$

Q69: The negation of the statement

$$\sim p \wedge (p \vee q)$$

(A) $\sim p \wedge \sim q$

(B) $\sim p \vee q$

(C) $\sim p \wedge q$

(D) $p \vee \sim q$

Q69: The negation of the statement

$$\sim p \wedge (p \vee q)$$

(A) $\sim p \wedge \sim q$

(B) $\sim p \vee q$

(C) $\sim p \wedge q$

(D) $p \vee \sim q$

Solution:

$$\begin{aligned} & \sim (\sim p \wedge (p \vee q)) \\ &= \sim ((\sim p \wedge p) \vee (\sim p \wedge q)) \\ &= \sim (\sim p \wedge q) = p \vee \sim q \end{aligned}$$

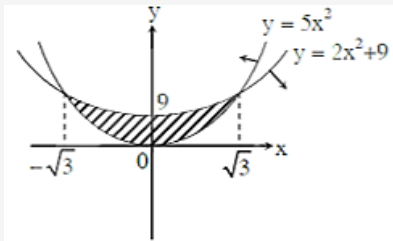
Q70: The area of the region: $R = \{(x, y) : 5x^2 \leq y \leq 2x^2 + 9\}$ is

- (A) $11\sqrt{3}$ square units
- (B) $12\sqrt{3}$ square units
- (C) $6\sqrt{3}$ square units
- (D) $9\sqrt{3}$ square units

Q70: The area of the region: $R = \{(x, y) : 5x^2 \leq y \leq 2x^2 + 9\}$ is

- (A) $11\sqrt{3}$ square units
- (B) $12\sqrt{3}$ square units
- (C) $6\sqrt{3}$ square units
- (D) $9\sqrt{3}$ square units

Solution:



Required area

$$\begin{aligned} &= 2 \int_0^{\sqrt{3}} (2x^2 + 9 - 5x^2) dx \\ &= 2 \int_0^{\sqrt{3}} (9 - 3x^2) dx \\ &= 2 \left[9x - x^3 \right]_0^{\sqrt{3}} = 12\sqrt{3} \end{aligned}$$

Q71: Let A and B be 3×3 real matrices such that A is symmetric matrix and B is skew-symmetric matrix. Then the system of linear equations $(A^2 B^2 - B^2 A^2) X = O$, where X is a 3×1 column matrix of unknown variables and O is a 3×1 null matrix, has

- (A) a unique solution
- (B) exactly two solutions
- (C) no solution
- (D) infinitely many solutions

Q71: Let A and B be 3×3 real matrices such that A is symmetric matrix and B is skew-symmetric matrix. Then the system of linear equations $(A^2 B^2 - B^2 A^2) X = O$, where X is a 3×1 column matrix of unknown variables and O is a 3×1 null matrix, has

- (A) a unique solution
- (B) exactly two solutions
- (C) no solution
- (D) infinitely many solutions

Solution:

$$A^T = A, B^T = -B$$

$$\text{Let } A^2 B^2 - B^2 A^2 = P$$

$$P^T = (A^2 B^2 - B^2 A^2)^T = (A^2 B^2)^T - (B^2 A^2)^T$$

$$= (B^2)^T (A^2)^T - (A^2)^T (B^2)^T$$

$$= B^2 A^2 - A^2 B^2$$

$$\Rightarrow P \text{ is skew-symmetric matrix}$$

$$\Rightarrow |P| = 0$$

Hence $PX = 0$ has infinite solutions

Q72: If P is a point on the parabola $y = x^2 + 4$ which is closest to the straight line $y = 4x - 1$, then the co-ordinates of P are

- (A) (3,13)
- (B) (2,8)
- (C) (-2,8)
- (D) (1,5)

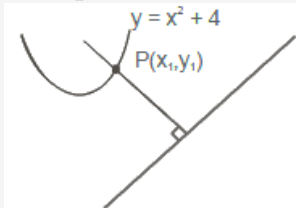
Q72: If P Is a point on the parabola $y = x^2 + 4$ which is closest to the straight line $y = 4x - 1$, then the co-ordinates of P are

- (A) (3,13)
- (B) (2,8)
- (C) (-2,8)
- (D) (1,5)

Solution:

$$\left. \frac{dy}{dx} \right|_P = 4$$

$$\therefore 2x_1 = 4$$



$$\Rightarrow x_1 = 2$$

$\therefore$ Point will be (2, 8)

Q73: For the statements p and q, consider the following compound statements :

(a) $(\sim q \wedge (p \rightarrow q)) \rightarrow \sim p$

(b) $((p \vee q) \wedge \sim p) \rightarrow q$

Then which of the following statements is correct ?

- (A) (a) and (b) both are tautologies.
- (B) (b) is a tautology but not (a).
- (C) (a) is a tautology but not (b).
- (D) (a) and (b) both are not tautologies.

Q73: For the statements p and q, consider the following compound statements :

(a) $(\sim q \wedge (p \rightarrow q)) \rightarrow \sim p$

(b) $((p \vee q) \wedge \sim p) \rightarrow q$

Then which of the following statements is correct ?

- (A) (a) and (b) both are tautologies.
- (B) (b) is a tautology but not (a).
- (C) (a) is a tautology but not (b).
- (D) (a) and (b) both are not tautologies.

Solution:

Truth table

p	q	$\sim p$	$\sim q$	$p \vee q$	$p \rightarrow q$	$\sim q \wedge (p \rightarrow q)$	$(p \vee q) \wedge \sim p$	(a)	(b)
T	T	F	F	T	T	F	F	T	T
T	F	F	T	T	F	F	F	T	T
F	T	T	F	T	T	F	T	T	T
F	F	T	T	F	T	T	F	T	T

Q74: Let $a, b \in \mathbb{R}$. If the mirror image of the point $P(a, 6, 9)$ with respect to the line $\frac{x-3}{7} = \frac{y-2}{5} = \frac{z-1}{-9}$ is $(20, b, -a - 9)$, then $|a + b|$ is equal to

- (A) 88
- (B) 84
- (C) 90
- (D) 86

Q74: Let $a, b \in \mathbb{R}$. If the mirror image of the point $P(a, 6, 9)$ with respect to the line $\frac{x-3}{7} = \frac{y-2}{5} = \frac{z-1}{-9}$ is $(20, b, -a - 9)$, then $|a + b|$ is equal to

(A) 88

(B) 84

(C) 90

(D) 86

Solution:

Mid – point of $P(a, 6, 9)$ and $(20, b, -a - 9)$ is $\left(\frac{a+20}{2}, \frac{6+b}{2}, \frac{-a}{2}\right)$ lies on the line

$$\text{So } \frac{\frac{a+20}{2}-3}{7} = \frac{\frac{6+b}{2}-2}{5} = \frac{\frac{-a}{2}-1}{-9}$$

$$\frac{a+14}{14} = \frac{2+b}{10} = \frac{a+2}{18} \Rightarrow b = -32, a = -56$$

$$\Rightarrow |a + b| = 88$$

Q75: The vector equation of the plane passing through the intersection of the planes

$\vec{r} \cdot (\hat{i} + \hat{j} + \hat{k}) = 1$ and $\vec{r} \cdot (\hat{i} - 2\hat{j}) = -2$, and the point $(1, 0, 2)$ is:

(A) $\vec{r} \cdot (\hat{i} + 7\hat{j} + 3\hat{k}) = 7$

(B) $\vec{r} \cdot (3\hat{i} + 7\hat{j} + 3\hat{k}) = 7$

(C) $\vec{r} \cdot (\hat{i} - 7\hat{j} + 3\hat{k}) = \frac{7}{3}$

(D) $\vec{r} \cdot (\hat{i} + 7\hat{j} + 3\hat{k}) = \frac{7}{3}$

Q75: The vector equation of the plane passing through the intersection of the planes

$\vec{r} \cdot (\hat{i} + \hat{j} + \hat{k}) = 1$ and $\vec{r} \cdot (\hat{i} - 2\hat{j}) = -2$, and the point $(1, 0, 2)$ is:

(A) $\vec{r} \cdot (\hat{i} + 7\hat{j} + 3\hat{k}) = 7$

(B) $\vec{r} \cdot (3\hat{i} + 7\hat{j} + 3\hat{k}) = 7$

(C) $\vec{r} \cdot (\hat{i} - 7\hat{j} + 3\hat{k}) = \frac{7}{3}$

(D) $\vec{r} \cdot (\hat{i} + 7\hat{j} + 3\hat{k}) = \frac{7}{3}$

Solution:

Plane passing through intersection of plane is

$$\left\{ \vec{r} \cdot (\hat{i} + \hat{j} + \hat{k}) - 1 \right\} + \lambda \left\{ \vec{r} \cdot (\hat{i} - 2\hat{j}) + 2 \right\} = 0$$

This plane passes through $\hat{i} + 2\hat{k}$, so we get

$$(3 - 1) + \lambda(1 + 2) = 0 \Rightarrow \lambda = -\frac{2}{3}$$

Hence, equation of plane is $3 \left\{ \vec{r} \cdot (\hat{i} + \hat{j} + \hat{k}) - 1 \right\} - 2 \left\{ \vec{r} \cdot (\hat{i} - 2\hat{j}) + 2 \right\} = 0$

$$\Rightarrow \vec{r} \cdot (\hat{i} + 7\hat{j} + 3\hat{k}) = 7$$

Q76: The angle of elevation of a jet plane from a point A on the ground is 60° . After a flight of 20 seconds at the speed of 432 km/hour, the angle of elevation changes to 30° . If the jet plane is flying at a constant height, then its height is :

(A) $2400\sqrt{3}\text{m}$

(B) $1800\sqrt{3}\text{m}$

(C) $1200\sqrt{3}\text{m}$

(D) $3600\sqrt{3}\text{m}$

Q76: The angle of elevation of a jet plane from a point A on the ground is 60° . After a flight of 20 seconds at the speed of 432 km/hour, the angle of elevation changes to 30° . If the jet plane is flying at a constant height, then its height is :

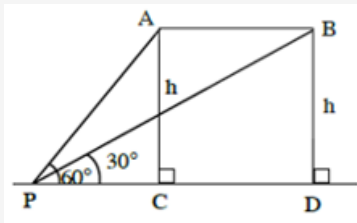
(A) $2400\sqrt{3}\text{m}$

(B) $1800\sqrt{3}\text{m}$

(C) $1200\sqrt{3}\text{m}$

(D) $3600\sqrt{3}\text{m}$

Solution:



$$v = 432 \times \frac{1000}{60 \times 60} \text{ m/sec} = 120 \text{ m/sec}$$

$$\text{Distance } AB = v \times 20 = 2400 \text{ meter}$$

In $\triangle PAC$

$$\tan 60^\circ = \frac{h}{PC} \Rightarrow PC = \frac{h}{\sqrt{3}}$$

In $\triangle PBD$

$$\tan 30^\circ = \frac{h}{PD} \Rightarrow PD = \sqrt{3}h$$

$$PD = PC + CD$$

$$\sqrt{3}h = \frac{h}{\sqrt{3}} + 2400 \Rightarrow \frac{2h}{\sqrt{3}} = 2400$$

$$h = 1200\sqrt{3} \text{ meter}$$

Q77: The probability that two randomly selected subsets of the set $\{1, 2, 3, 4, 5\}$ have exactly two elements in their intersection, is :

(A) $\frac{65}{2^8}$

(B) $\frac{35}{2^7}$

(C) $\frac{135}{2^9}$

(D) $\frac{65}{2^7}$

Q77: The probability that two randomly selected subsets of the set $\{1, 2, 3, 4, 5\}$ have exactly two elements in their intersection, is :

(A) $\frac{65}{2^8}$

(B) $\frac{35}{2^7}$

(C) $\frac{135}{2^9}$

(D) $\frac{65}{2^7}$

Solution:

$$\begin{aligned} \text{Required probability} &= \frac{{}^5C_2 \times 3^3}{4^5} \\ &= \frac{10 \times 27}{2^{10}} = \frac{135}{2^9} \end{aligned}$$

Q78: If $n \geq 2$ is a positive integer, then the sum of the series ${}^{n+1}C_2 + 2({}^2C_2 + {}^3C_2 + {}^4C_2 + \dots + {}^nC_2)$ is

(A) $\frac{n(n+1)^2(n+2)}{12}$

(B) $\frac{n(n+1)(2n+1)}{6}$

(C) $\frac{n(n-1)(2n+1)}{6}$

(D) $\frac{n(2n+1)(3n+1)}{6}$

Q78: If $n \geq 2$ is a positive integer, then the sum of the series ${}^{n+1}C_2 + 2({}^2C_2 + {}^3C_2 + {}^4C_2 + \dots + {}^nC_2)$ is

(A) $\frac{n(n+1)^2(n+2)}{12}$

(B) $\frac{n(n+1)(2n+1)}{6}$

(C) $\frac{n(n-1)(2n+1)}{6}$

(D) $\frac{n(2n+1)(3n+1)}{6}$

Solution:

$$\begin{aligned} S &= {}^2C_2 + {}^3C_2 + \dots + {}^nC_2 = {}^{n+1}C_3 \\ \therefore {}^{n+1}C_2 + {}^{n+1}C_3 + {}^{n+1}C_3 &= {}^{n+2}C_3 + {}^{n+1}C_3 \\ &= \frac{(n+2)!}{3!(n-1)!} + \frac{(n+1)!}{3!(n-2)!} \\ &= \frac{(n+2)(n+1)n}{6} + \frac{(n+1)(n)(n-1)}{6} = \frac{n(n+1)}{6}(2n+1) \end{aligned}$$

Q79: If a curve $y = f(x)$ passes through the point (1,2) and satisfies $x \frac{dy}{dx} + y = bx^4$, then for what value of b, $\int_1^2 f(x) dx = \frac{62}{5}$?

- (A) 10
- (B) 5
- (C) 31/5
- (D) 62/5

Q79: If a curve $y = f(x)$ passes through the point (1,2) and satisfies $x \frac{dy}{dx} + y = bx^4$, then for what value of b, $\int_1^2 f(x) dx = \frac{62}{5}$?

(A) 10

(B) 5

(C) 31/5

(D) 62/5

Solution:

$$\frac{dy}{dx} + \frac{y}{x} = bx^3$$

$$\text{I.F.} = e^{\int \frac{dx}{x}} = x$$

$$\therefore yx = \int bx^4 dx = \frac{bx^5}{5} + C$$

Passes through (1, 2), we get

$$2 = \frac{b}{5} + C \quad \dots (i)$$

$$\text{Also, } \int_1^2 \left(\frac{bx^4}{5} + \frac{C}{x} \right) dx = \frac{62}{5}$$

$$\Rightarrow \frac{b}{25} \times 32 + C \ln 2 - \frac{b}{25} = \frac{62}{5}$$

$$\Rightarrow C = 0 \text{ and } b = 10$$

Q80: If the curve $y = ax^2 + bx + c$, $x \in \mathbb{R}$, passes through the point (1, 2) and the tangent line to this curve at origin is $y = x$, then the possible values of a , b , c are :

- (A) $a = 1, b = 1, c = 0$
- (B) $a = \frac{1}{2}, b = \frac{1}{2}, c = 1$
- (C) $a = -1, b = 1, c = 1$
- (D) $a = 1, b = 0, c = 1$

Q80: If the curve $y = ax^2 + bx + c$, $x \in \mathbb{R}$, passes through the point (1, 2) and the tangent line to this curve at origin is $y = x$, then the possible values of a , b , c are :

- (A) $a = 1, b = 1, c = 0$
- (B) $a = \frac{1}{2}, b = \frac{1}{2}, c = 1$
- (C) $a = -1, b = 1, c = 1$
- (D) $a = 1, b = 0, c = 1$

Solution:

$$\begin{aligned} 2 &= a + b + c \quad \dots (i) \\ \frac{dy}{dx} &= 2ax + b \Rightarrow \left. \frac{dy}{dx} \right|_{(0,0)} = 1 \\ \Rightarrow b &= 1 \Rightarrow a + c = 1 \end{aligned}$$

Q81: If $a + \alpha = 1$, $b + \beta = 2$ and $af(x) + \alpha f\left(\frac{1}{x}\right) = bx + \frac{\beta}{x}$, $x \neq 0$, then the value of the expression $\frac{f(x)+f\left(\frac{1}{x}\right)}{x+\frac{1}{x}}$ is _____.

Q81: If $a + \alpha = 1$, $b + \beta = 2$ and $af(x) + \alpha f\left(\frac{1}{x}\right) = bx + \frac{\beta}{x}$, $x \neq 0$, then the value of the expression $\frac{f(x)+f\left(\frac{1}{x}\right)}{x+\frac{1}{x}}$ is _____.

2.00

Solution:

$$af(x) + \alpha f\left(\frac{1}{x}\right) = bx + \frac{\beta}{x} \quad \dots (i)$$

$$x \rightarrow \frac{1}{x}$$

$$af\left(\frac{1}{x}\right) + \alpha f(x) = \frac{b}{x} + \beta x \quad \dots (ii)$$

$$(i) + (ii)$$

$$(a + \alpha) \left[f(x) + f\left(\frac{1}{x}\right) \right] = \left(x + \frac{1}{x} \right) (b + \beta)$$

$$\frac{f(x)+f\left(\frac{1}{x}\right)}{x+\frac{1}{x}} = \frac{2}{1} = 2$$

Q82: Let a point P be such that its distance from the point (5, 0) is thrice the distance of P from the point (−5, 0). If the locus of the point P is a circle of radius r, then $4r^2$ is equal to _____.

(Round off the answer to nearest integer)

Q82: Let a point P be such that its distance from the point (5, 0) is thrice the distance of P from the point (−5, 0). If the locus of the point P is a circle of radius r, then $4r^2$ is equal to _____.

(Round off the answer to nearest integer)

56.00

Solution:

Internal point which divides (5, 0) and (−5, 0) in the ratio 3 : 1 is $\left(\frac{-5}{2}, 0\right)$

And external point which divides (5, 0) and (−5, 0) in the ratio 3 : 1 is (−10, 0)

$$2r = \left(\frac{-5}{2} + 10\right) = \frac{15}{2} = 7.5$$

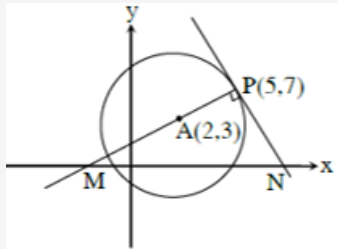
$$(2r)^2 = 56.25 \approx 56$$

Q83: If the area of the triangle formed by the positive x-axis, the normal and the tangent to the circle $(x - 2)^2 + (y - 3)^2 = 25$ at the point $(5, 7)$ is A , then $24A$ is equal to

Q83: If the area of the triangle formed by the positive x-axis, the normal and the tangent to the circle $(x - 2)^2 + (y - 3)^2 = 25$ at the point $(5, 7)$ is A , then $24A$ is equal to

1225.00

Solution:



Equation of normal at P

$$(y - 7) = \left(\frac{7 - 3}{5 - 2} \right) (x - 5)$$

$$3y - 21 = 4x - 20$$

$$\Rightarrow 4x - 3y + 1 = 0 \quad \dots (i)$$

$$\Rightarrow M \left(-\frac{1}{4}, 0 \right)$$

Equation of tangent at P

$$(y - 7) = -\frac{3}{4}(x - 5)$$

$$4y - 28 = -3x + 15$$

$$\Rightarrow 3x + 4y = 43 \quad \dots (ii)$$

$$\Rightarrow N \left(\frac{43}{3}, 0 \right)$$

$$\text{Hence, } ar(\triangle PMN) = \frac{1}{2} \times MN \times 7$$

$$A = \frac{1}{2} \times \frac{175}{12} \times 7$$

$$\Rightarrow 24A = 1225$$

(In the question positive x – axis is given which do not from any triangle for required area)

Q84: Let $i = \sqrt{-1}$. If $\frac{(-1+i\sqrt{3})^{21}}{(1-i)^{24}} + \frac{(1+i\sqrt{3})^{21}}{(1+i)^{24}} = k$, and $n = [|k|]$ be the greatest integral part of $|k|$. Then $\sum_{j=0}^{n+5} (j+5)^2 - \sum_{j=0}^{n+5} (j+5)$ is equal to _____.

Q84: Let $i = \sqrt{-1}$. If $\frac{(-1+i\sqrt{3})^{21}}{(1-i)^{24}} + \frac{(1+i\sqrt{3})^{21}}{(1+i)^{24}} = k$, and $n = [|k|]$ be the greatest integral part of $|k|$. Then $\sum_{j=0}^{n+5} (j+5)^2 - \sum_{j=0}^{n+5} (j+5)$ is equal to _____.

310.00

Solution:

$$\begin{aligned} \frac{(-1+i\sqrt{3})^{21}}{(-2i)^{12}} + \frac{(1+i\sqrt{3})^{21}}{(2i)^{12}} &= k \\ \Rightarrow \frac{1}{2^{12}} \left[(1+i\sqrt{3})^{21} - (1-i\sqrt{3})^{21} \right] &= k \\ \Rightarrow \frac{1}{2^{12}} \left[2^{21} \left(\cos \frac{\pi}{3} + i \sin \frac{\pi}{3} \right)^{21} - 2^{21} \left(\cos \frac{\pi}{3} - i \sin \frac{\pi}{3} \right)^{21} \right] &= k \\ \Rightarrow i 2^9 \left(2 \sin \frac{21\pi}{3} \right) &= k \\ \Rightarrow k &= 0 \end{aligned}$$

Now $n = [|x|] = 0$

$$\text{Then } \sum_{j=0}^5 (j+5)^2 - \sum_{j=0}^5 (j+5) = \sum_{j=0}^5 (j+5)^2 - (j+5)$$

$$\sum_{j=0}^5 (j^2 + 9j + 20) = \left(\sum_{j=1}^5 j^2 + 9 \sum_{j=1}^5 j + 20 \sum_{j=1}^5 1 \right)$$

$$= 310$$

Q85: The number of the real roots of the equation $(x + 1)^2 + |x - 5| = \frac{27}{4}$ is

Q85: The number of the real roots of the equation $(x + 1)^2 + |x - 5| = \frac{27}{4}$ is

2.00

Solution:

$$(i) \ x \geq 5$$

$$x^2 + 3x - 4 = \frac{27}{4}$$

$$4x^2 + 12x - 43 = 0$$

$$D \geq 0$$

$$x = \frac{-12 \pm \sqrt{144 + 16 \times 43}}{8}$$

$$\text{For } x = \frac{-3 \pm \sqrt{52}}{2}$$

For $x \geq 5$, both not possible

$$(ii) \ x < 5$$

$$x^2 + x + 5 = \frac{27}{4}$$

$$4x^2 + 4x - 3 = 0$$

$$x = \frac{-4 \pm \sqrt{16 + 48}}{8}$$

$$x = \frac{1}{2}, \ x = \frac{-3}{2} \text{ both possible for } x < 5$$

Q86: If the variance of 10 natural numbers 1, 1, 1, ..., 1, k is less than 10, then the maximum possible value of k is _____.

Q86: If the variance of 10 natural numbers 1, 1, 1, ..., 1, k is less than 10, then the maximum possible value of k is _____.

11.00

Solution:

$$\sigma^2 = \frac{\sum x^2}{n} - \left(\frac{\sum x}{n} \right)^2$$

$$\sigma^2 = \frac{(9+k)^2}{10} - \left(\frac{9+k}{10} \right)^2 < 10$$

$$(9 + k^2) 10 - (81 + k^2 + 18k) < 1000$$

$$90 + 10k^2 - k^2 - 18k - 81 \leq 1000$$

$$9k^2 - 18k + 9 < 1000$$

$$(k - 1)^2 < \frac{1000}{9} \Rightarrow k - 1 < \frac{10\sqrt{10}}{3}$$

$$k < \frac{10\sqrt{10}}{3} + 1$$

Maximum integral value of $k = 11$

Q87: Let λ be an integer. If the shortest distance between the lines $x - \lambda = 2y - 1 = -2z$ and $x = y + 2\lambda = z - \lambda$ is $\frac{\sqrt{7}}{2\sqrt{2}}$, then the value of $|\lambda|$ is ____.

Q87: Let λ be an integer. If the shortest distance between the lines $x - \lambda = 2y - 1 = -2z$ and $x = y + 2\lambda = z - \lambda$ is $\frac{\sqrt{7}}{2\sqrt{2}}$, then the value of $|\lambda|$ is ____.

1.00

Solution:

$$L_1 : \frac{x-\lambda}{1} = \frac{y-\frac{1}{2}}{\frac{1}{2}} = \frac{z}{-\frac{1}{2}} \text{ and}$$

$$L_2 : \frac{x}{1} = \frac{y+2\lambda}{1} = \frac{z-\lambda}{1}$$

$$D = \frac{\left| (\vec{a}_1 - \vec{a}_2) \cdot (\vec{b}_1 \times \vec{b}_2) \right|}{\left| \vec{b}_1 \times \vec{b}_2 \right|}$$

$$\vec{b}_1 \times \vec{b}_2 = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 1 & \frac{1}{2} & -\frac{1}{2} \\ 1 & 1 & 1 \end{vmatrix}$$

$$= \hat{i}(1) - \hat{j}\left(\frac{3}{2}\right) + \hat{k}\left(\frac{1}{2}\right)$$

$$= \hat{i} - \frac{3\hat{j}}{2} + \frac{\hat{k}}{2}$$

$$\left| \vec{b}_1 \times \vec{b}_2 \right| = \sqrt{1 + \frac{9}{4} + \frac{1}{4}} = \sqrt{\frac{7}{2}}$$

$$(\vec{a}_1 - \vec{a}_2) = \lambda\hat{i} + \left(\frac{1}{2} + 2\lambda\right)\hat{j} - \lambda\hat{k}$$

$$D = \frac{\left| \lambda - \left(\frac{3}{4} + 3\lambda\right) - \frac{\lambda}{2} \right|}{\sqrt{\frac{7}{2}}} = \sqrt{\frac{7}{2}} \times \frac{1}{2}$$

$$\left| \frac{5\lambda}{2} + \frac{3}{4} \right| = \frac{7}{4} \Rightarrow \lambda = \frac{2}{5}, \lambda = -1$$

As $\lambda \in \text{integer}$, so $|\lambda| = 1$

Q88: The students $S_1, S_2, \dots, S_{10}$ are to be divided into 3 groups. A, B and C such that each group has at least one student and the group C has at most 3 students. Then the total number of possibilities of forming such groups is ____.

Q88: The students $S_1, S_2, \dots, S_{10}$ are to be divided into 3 groups. A, B and C such that each group has at least one student and the group C has at most 3 students. Then the total number of possibilities of forming such groups is _____.

31650.00

Solution:

A	B	C
1	8	1
2	7	1
$\vdots$	$\vdots$	$\vdots$
6	1	3

$$\begin{aligned}
 \text{Ways to distribute in groups} &= {}^{10}C_1 ({}^9C_1 + \dots + {}^9C_8) + {}^{10}C_2 ({}^8C_1 + \dots + {}^8C_7) + {}^{10}C_3 ({}^7C_1 + \dots + {}^7C_6) \\
 &= 10(510) + 45(254) + 120(126) \\
 &= 31650
 \end{aligned}$$

Q89: The sum of first four terms of a geometric progression (G.P.) is $\frac{65}{12}$ and the sum of their respective reciprocals is $\frac{65}{18}$. If the product of first three terms of the G.P. is 1, and the third term is α , then 2α is _____.

Q89: The sum of first four terms of a geometric progression (G.P.) is $\frac{65}{12}$ and the sum of their respective reciprocals is $\frac{65}{18}$. If the product of first three terms of the G.P. is 1, and the third term is α , then 2α is _____.

3.00

Solution:

Let terms be a, ar, ar^2, ar^3

$$a + ar + ar^2 + ar^3 = \frac{65}{12} \quad \dots (i)$$

$$\frac{1}{a} + \frac{1}{ar} + \frac{1}{ar^2} + \frac{1}{ar^3} = \frac{65}{18}$$

$$\frac{1}{a} \left(\frac{r^3 + r^2 + r + 1}{r^3} \right) = \frac{65}{18} \quad \dots (ii)$$

$$\frac{(i)}{(ii)}, \quad a^2 r^3 = \frac{18}{12} = \frac{3}{2}$$

$$a^3 r^3 = 1 \Rightarrow a \left(\frac{3}{2} \right) = 1 \Rightarrow a = \frac{2}{3}$$

$$\frac{4}{9} r^3 = \frac{3}{2} \Rightarrow r^3 = \frac{3^3}{2^3} \Rightarrow r = \frac{3}{2}$$

$$\alpha = ar^2 = \frac{2}{3} \cdot \left(\frac{3}{2} \right)^2 = \frac{3}{2}$$

$$2\alpha = 3$$

Q90: For integers n and r, let $\binom{n}{r} = \begin{cases} {}^nC_r, & \text{if } n \geq r \geq 0 \\ 0, & \text{otherwise} \end{cases}$

The value of k for which the sum

$\sum_{i=0}^k \binom{10}{i} \binom{15}{k-i} + \sum_{i=0}^{k+1} \binom{12}{i} \binom{13}{k+1-i}$ is maximum, is equal to ____.

Q90: For integers n and r, let $\binom{n}{r} = \begin{cases} {}^nC_r, & \text{if } n \geq r \geq 0 \\ 0, & \text{otherwise} \end{cases}$

The value of k for which the sum

$\sum_{i=0}^k \binom{10}{i} \binom{15}{k-i} + \sum_{i=0}^{k+1} \binom{12}{i} \binom{13}{k+1-i}$ is maximum, is equal to ____.

12.00

Solution:

$$\begin{aligned} & \sum_{i=0}^k \binom{10}{i} \binom{15}{k-i} \\ &= {}^{10}C_0 \cdot {}^{15}C_k + {}^{10}C_1 \cdot {}^{15}C_{k-1} + {}^{10}C_2 \cdot {}^{15}C_{k-2} + \dots + {}^{10}C_k \cdot {}^{15}C_0 \\ (1+x)^{10} &= {}^{10}C_0 + {}^{10}C_1 x + {}^{10}C_2 x^2 + {}^{10}C_3 x^3 + \dots + {}^{10}C_k x^k + \dots \\ (1+x)^{15} &= {}^{15}C_0 + {}^{15}C_1 x + {}^{15}C_2 x^2 + \dots + {}^{15}C_k x^k + \dots \end{aligned}$$

Now

$$(1+x)^{10} \cdot (1+x)^{15} = ()()$$

Compare coefficient of x^k both side

$${}^{25}C_k = {}^{10}C_0 \cdot {}^{15}C_k + \dots + {}^{10}C_k \cdot {}^{15}C_0$$

$$\text{Similarly, } \sum_{i=0}^{k+1} \binom{12}{i} \binom{13}{k+1-i} = {}^{25}C_{k+1}$$

$$\text{Now } {}^{25}C_k + {}^{25}C_{k+1}$$

$$= {}^{26}C_{k+1}$$

This is maximum for $k = 12$

